

Aristotle University of Thessaloniki
Department of Electrical and Computer Engineering
Telecommunications Division

Konstantinos Manolakis 11289

Communication Systems II

Constellations Dataset and VLM-CNN Evaluation and Comparison

1 Part I: IQ Signal Modeling and Dataset Construction

1.1 Signal Generation and Modulation Schemes

All signals were generated using MATLAB whilst also using the communications toolbox, in complex baseband form, where each transmitted symbol is represented as a complex number $s = I + jQ$, with I and Q denoting the in-phase and quadrature components respectively. The baseband representation eliminates the need to model the carrier frequency explicitly, as the transmitted passband signal is given by:

$$x(t) = I \cdot \cos(2\pi f_c t) - Q \cdot \sin(2\pi f_c t) \quad (1)$$

The following modulation schemes were implemented, as specified in the projects' instructions: 4-ASK, 8-ASK, BPSK, QPSK, 4-HQAM, 16-HQAM, 64-HQAM, 16-QAM, 32-QAM, 64-QAM, 128-QAM, 256-QAM, 16-APSK, 32-APSK, 64-APSK, and 128-APSK, totalling 16 distinct modulation schemes.

1.1.1 Quadrature Amplitude Modulation (QAM)

For square M -QAM constellations where $\log_2 M$ is even, symbols are drawn from a two-dimensional grid with in-phase and quadrature components taking values in the set $\{-(m-1), -(m-3), \dots, +(m-1)\}$, where $m = \sqrt{M}$. For non square M -QAM constellations the approach is similar but the grid is orthogonal instead of square. For the constellations of QAM modulations, the "qammod" function from the communications toolbox was used and the "awgn" function was used to add Additive White Gaussian Noise.

1.1.2 Amplitude Shift Keying (ASK)

M -ASK encodes information exclusively in the signal amplitude, with symbols confined to the in-phase axis ($Q = 0$). The symbol alphabet is identical to a one-dimensional PAM constellation: $s \in \{-(M-1), -(M-3), \dots, +(M-1)\}$. Although the terms PAM and ASK are often used interchangeably in the baseband context, they refer to the same underlying one-dimensional constellation. For the constellations of ASK modulations, the "pammod" function from the communications toolbox was used whilst also adding Additive White Gaussian Noise with the "awgn" function.

1.1.3 Hexagonal QAM (HQAM)

Unlike conventional square QAM, HQAM places constellation points on a hexagonal lattice, which achieves a denser packing in the two-dimensional signal space. The lattice basis vectors are:

$$\mathbf{e}_1 = (1, 0), \quad \mathbf{e}_2 = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \quad (2)$$

Each lattice point is defined as $p(m, n) = m\mathbf{e}_1 + n\mathbf{e}_2$, and the M points closest to the origin (excluding the origin itself) are selected to form the constellation. This results in a more uniform distribution of inter-symbol distances compared to square QAM, offering a theoretical packing gain. This type of modulation is showing significant promise in next generation telecommunication systems (6g). The algorithm that was used to create the H-QAM constellations was the following:

```

function [tx, rx] = gen_hqam(M, N, snr)
    v1 = 1; v2 = 0.5 + 1j*sqrt(3)/2;
    gs = ceil(sqrt(M)) + 2;
    [mg,ng] = meshgrid(-gs:gs,-gs:gs);
    gp = mg(:)*v1 + ng(:)*v2;
    gp = gp(abs(gp) > 1e-6);
    [~,idx] = sort(abs(gp));
    hc = gp(idx(1:M));
    hc = hc / sqrt(mean(abs(hc).^2));
    tx = hc(randi([1,M], 1, N)).';
    rx = awgn(tx, snr, 'measured');
end

```

1.1.4 Amplitude and Phase Shift Keying (APSK)

APSK arranges constellation points on concentric rings, encoding information in both amplitude and phase. For M -APSK with n rings, the k -th point on the i -th ring is given by:

$$s_k = r_i \cdot e^{j\theta_k}, \quad \theta_k = \frac{2\pi k}{n_i} + \phi_i \quad (3)$$

where r_i is the radius of the i -th ring, n_i is the number of points on that ring, and ϕ_i is the initial phase offset. The ring radii ratios were selected according to the DVB-S2 standard [1], which optimizes the minimum Euclidean distance between constellation points. The configurations used are summarized in Table 1.

For the constellations generation the communications toolbox was also utilized, specifically the "apskmod" and "awgn" functions

Table 1: APSK ring configurations (DVB-S2 standard)

Scheme	Points per ring	Radii ratios
16-APSK	4 + 12	$\gamma = 2.57$
32-APSK	4 + 12 + 16	$\gamma_1 = 2.53, \gamma_2 = 4.30$
64-APSK	4 + 12 + 20 + 28	$\gamma_1 = 2.57, \gamma_2 = 4.62, \gamma_3 = 6.88$
128-APSK	4 + 12 + 20 + 40 + 52	$\gamma_1 = 2.53, \gamma_2 = 4.33, \gamma_3 = 6.93, \gamma_4 = 9.95$

1.1.5 Phase Shift Keying (PSK)

Phase Shift Keying encodes information exclusively in the phase of the transmitted signal, while the amplitude remains constant. All constellation points lie on the unit circle, so PSK symbols are inherently normalized to unit energy without requiring further processing.

For M -PSK, the k -th symbol is given by:

$$s_k = e^{j\left(\frac{2\pi k}{M} + \phi_0\right)}, \quad k = 0, 1, \dots, M-1 \quad (4)$$

where ϕ_0 is an initial phase offset. Two specific cases were implemented:

BPSK ($M = 2$): Binary Phase Shift Keying uses two antidiagonal symbols on the real axis, $s \in \{-1, +1\}$, encoding 1 bit per symbol.

QPSK ($M = 4$): Quadrature Phase Shift Keying uses four symbols equally spaced on the unit circle with a phase offset of $\phi_0 = \pi/4$, yielding $s \in \left\{\frac{1+j}{\sqrt{2}}, \frac{-1+j}{\sqrt{2}}, \frac{-1-j}{\sqrt{2}}, \frac{1-j}{\sqrt{2}}\right\}$, encoding 2 bits

per symbol. The $\pi/4$ offset ensures that no symbol transition passes through the origin, reducing the peak-to-average power ratio compared to standard QPSK. For the constellations generation the communications toolbox was utilized, specifically the "pskmod" and "awgn" functions

1.2 Energy Normalization

All generated symbol sequences were normalized to unit average energy prior to the addition of any impairments:

$$s_{\text{norm}} = \frac{s}{\sqrt{E_s}}, \quad E_s = \mathbb{E}[|s|^2] = \frac{1}{N} \sum_{i=1}^N |s_i|^2 \quad (5)$$

This normalization ensures that the signal-to-noise ratio is consistently defined across all modulation schemes, enabling fair comparison. After normalization, $\mathbb{E}[|s_{\text{norm}}|^2] = 1$ for all schemes, so that the noise power is determined solely by the target SNR value.

1.3 Transceiver Impairment Modeling

Four types of transceiver impairments were modeled directly in the IQ domain. Each impairment was defined at four discrete severity levels: *none*, *mild*, *medium*, and *severe*.

1.3.1 Additive White Gaussian Noise (AWGN)

Additive white Gaussian noise was applied to all signals. The received signal is:

$$r = s + n, \quad n \sim \mathcal{N}(0, \sigma_n^2) \quad (6)$$

where $\sigma_n^2 = 1/\text{SNR}_{\text{linear}}$ after energy normalization. The SNR in dB is defined as:

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \left(\frac{E_s}{\sigma_n^2} \right) \quad (7)$$

Eight discrete SNR levels were used: $\{0, 5, 10, 15, 20, 25, 30, 35\}$ dB. With the severity levels labeling philosophy listed below:

Table 2: SNR ranges and corresponding dataset labels

SNR (dB)	Condition	Label
0, 5	$\text{SNR} \leq 5$	very_low
10	$5 < \text{SNR} \leq 10$	low
15, 20	$10 < \text{SNR} \leq 20$	medium
25, 30	$20 < \text{SNR} \leq 30$	high
35	$\text{SNR} > 30$	very_high

1.3.2 Phase Noise

Phase noise is the random fluctuation of a signal's phase in oscillators and radio frequency systems, causing spectral spreading around the carrier frequency. Phase noise was modeled as a wrapped Gaussian random process applied to each symbol independently:

$$r = s \cdot e^{j\phi}, \quad \phi = \text{wrap}(\mathcal{N}(0, \sigma_\phi^2)) \quad (8)$$

where the wrapping operation constrains $\phi \in [-\pi, \pi)$. The severity levels and corresponding σ_ϕ values are listed in Table 3.

1.3.3 I/Q Imbalance

I/Q imbalance is a mismatch in amplitude or phase between the in-phase (I) and quadrature (Q) signal paths in a communication system. I/Q imbalance was modeled using an amplitude imbalance g and a phase imbalance $\Delta\phi$ between the in-phase and quadrature branches:

$$r = I_{\text{out}} + jQ_{\text{out}} \quad (9)$$

$$I_{\text{out}} = I \cos \frac{\Delta\phi}{2} - gQ \sin \frac{\Delta\phi}{2}, \quad Q_{\text{out}} = I \sin \frac{\Delta\phi}{2} + gQ \cos \frac{\Delta\phi}{2} \quad (10)$$

where $g = 10^{g_{\text{dB}}/20}$ is the linear amplitude imbalance factor.

1.3.4 Amplitude Distortion

Amplitude distortion is the unwanted alteration of a signal's amplitude, causing it to deviate from its intended waveform due to nonlinearities or limitations in system components. Amplitude distortion was modeled using a memoryless soft-limiter, which captures the AM/AM nonlinearity of a saturating power amplifier:

$$g(r) = \frac{|s|}{\sqrt{1 + (|s|/A_{\text{sat}})^2}} \quad (11)$$

where $A_{\text{sat}} = 10^{\text{IBO}_{\text{dB}}/20}$ is the saturation amplitude and IBO denotes the Input Back-Off from the compression point. The distorted signal is $r = g(|s|) \cdot e^{j\angle s}$, followed by re-normalization to unit power.

1.3.5 External Interference (Jamming)

External interference was modeled as a narrowband jammer consisting of a tone at a random phase, added to the received signal:

$$r = s + n + j_{\text{amp}} \cdot e^{j\psi}, \quad \psi \sim \mathcal{U}[0, 2\pi) \quad (12)$$

where $j_{\text{amp}} = \sqrt{J \cdot P_s}$ and $J = 10^{\text{JNR}_{\text{dB}}/10}$ is the jammer-to-signal power ratio.

Table 3: Impairment severity levels

Impairment	Parameter	None	Mild	Medium	Severe
Phase Noise	σ_ϕ (rad)	0	0.02	0.05	0.15
I/Q Imbalance	g (dB)	0	0.5	1.0	2.0
	$\Delta\phi$ (deg)	0	2	5	10
Amp. Distortion	IBO (dB)	∞	10	5	2
Jamming	JNR (dB)	$-\infty$	-10	0	+10

1.4 Constellation Image Generation

Firstly, the signals were created using a conventional Cartesian (I,Q) scatter diagram with stars symbolizing the ideal transmitted signal for each modulation type and the blue dots the received signal with all the impairments and the additive gaussian noise. It is important to mention that each constellation scatter diagram consists of 2000 received signals. One example is listed in Figure 1

Although the first iteration of the dataset was great to look at, it wasn't very useful in the context of training Convolutional Neural Networks (CNNs) and Vision Language Models

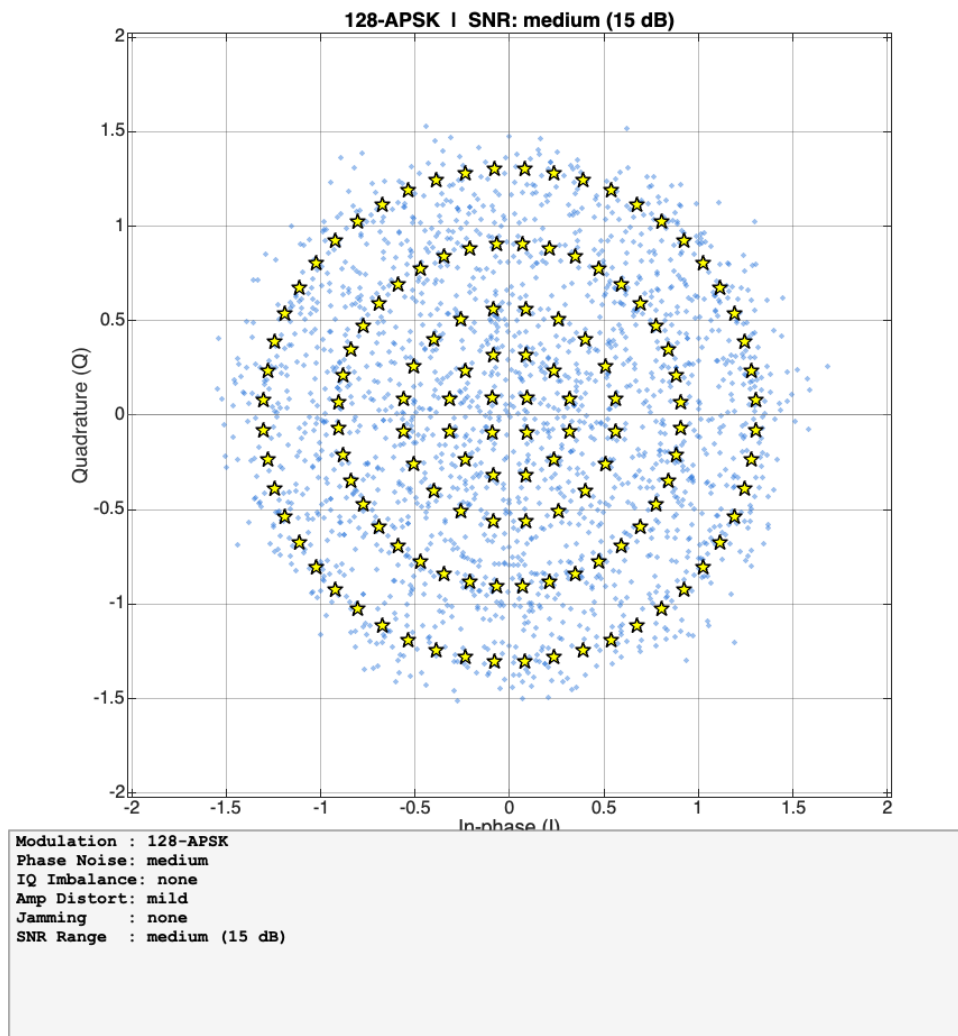


Figure 1: 128-APSK modulation with severity levels listed in the legend.

(VLMs). To create the final dataset each received signal was converted to a constellation image using a polar coordinate representation. Rather than plotting the conventional Cartesian (I, Q) scatter diagram, each symbol $s = I + jQ$ was mapped to polar form (r, θ) :

$$r = |s|, \quad \theta = \angle s \in [-\pi, \pi) \quad (13)$$

and displayed as a two-dimensional scatter plot with r on the horizontal axis and θ on the vertical axis. This representation offers two advantages over the Cartesian form. First, the radial structure of APSK constellations maps to horizontal bands, making the ring structure immediately apparent. Second, phase-related distortions such as phase noise manifest as vertical spreading, while amplitude distortions manifest as horizontal spreading, providing a more disentangled visual encoding of impairment types. That approach makes it easier for the models to understand patterns thus making it a logical approach, as shown in [2].

All images were rendered without axes, tick marks, grid lines, or titles, to remove unnecessary information and saved as 256×256 grayscale PNG files. This ensures that the neural networks trained on this dataset extract features exclusively from the geometric distribution of the scatter points rather than from annotation artifacts.

1.5 Dataset Structure and Labels

The final dataset consists of 32,768 images, generated by exhaustively combining all modulation schemes, SNR levels, and impairment severity levels:

$$16 \text{ modulations} \times 8 \text{ SNR levels} \times 4^4 \text{ impairment combinations} = 32,768 \text{ images} \quad (14)$$

Each image is associated with a structured label vector answering five classification questions:

1. **Modulation type:** one of 16 classes
2. **Phase noise severity:** {none, mild, medium, severe}
3. **I/Q imbalance severity:** {none, mild, medium, severe}
4. **Jamming severity:** {none, mild, medium, severe}
5. **SNR range:** {very_low, low, medium, high, very_high}

Labels are stored in a CSV file alongside the corresponding image filenames, enabling flexible multi-task training and evaluation. An example of the images used for model training are shown in Figure 2.

1.6 Symbol Error Probability vs SNR

Monte Carlo simulations were conducted to evaluate the Symbol Error Probability (SEP) as a function of SNR for QAM constellations of orders $M \in \{16, 32, 64, 128, 256\}$, both with and without phase noise. For square QAM constellations ($M = 16, 64, 256$), the theoretical SEP in the absence of phase noise is given by the closed-form expression [3]:

$$P_e \approx 4 \left(1 - \frac{1}{\sqrt{M}}\right) Q\left(\sqrt{\frac{3 \text{SNR}}{M-1}}\right) - 4 \left(1 - \frac{1}{\sqrt{M}}\right)^2 Q^2\left(\sqrt{\frac{3 \text{SNR}}{M-1}}\right) \quad (15)$$

where $Q(x) = \frac{1}{2} \text{erfc}(x/\sqrt{2})$. For non-square constellations ($M = 32, 128$), no closed-form expression exists; a nearest-neighbor approximation based on the actual MATLAB constellation geometry was used instead:

$$P_e \approx \bar{N} \cdot Q\left(\frac{d_{\min}}{\sqrt{2No}}\right) \quad (16)$$

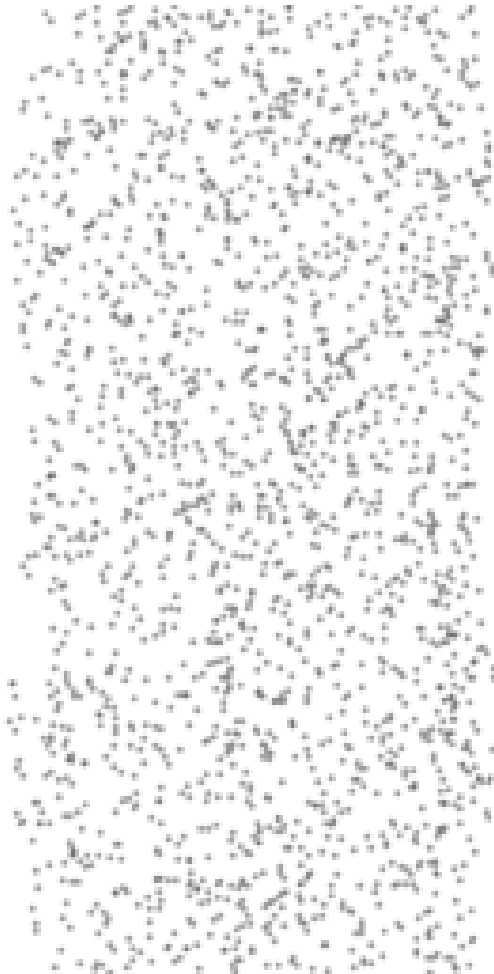
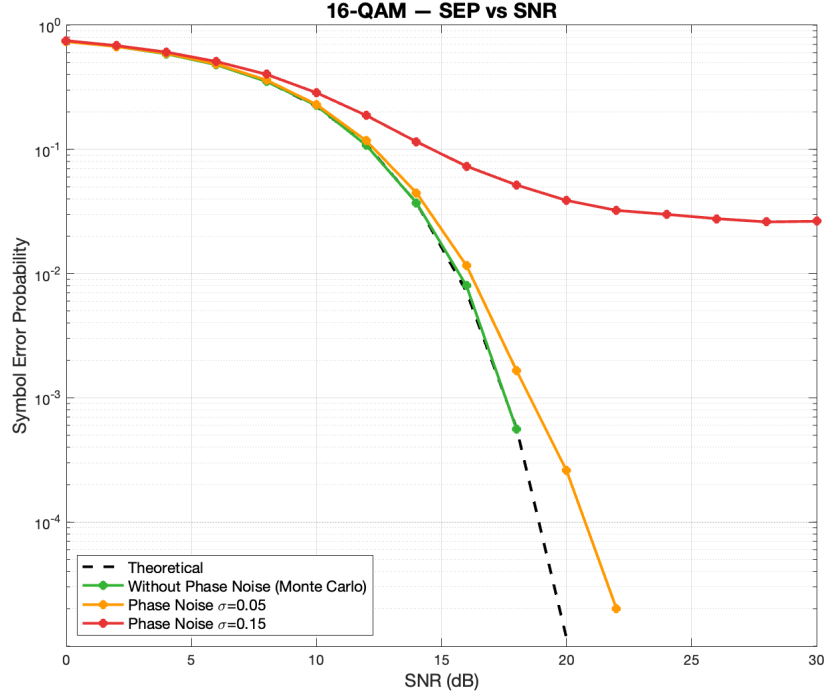
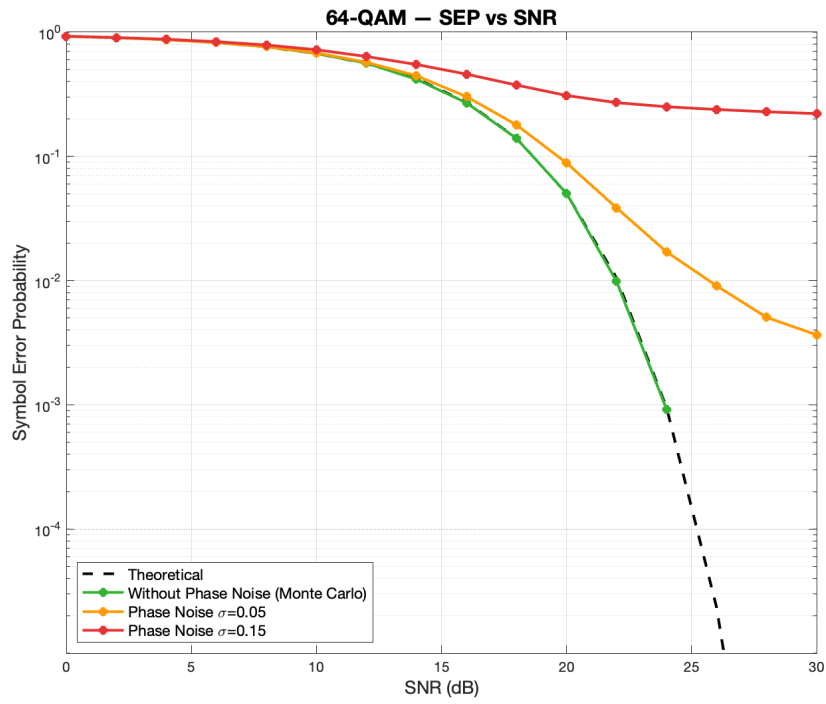
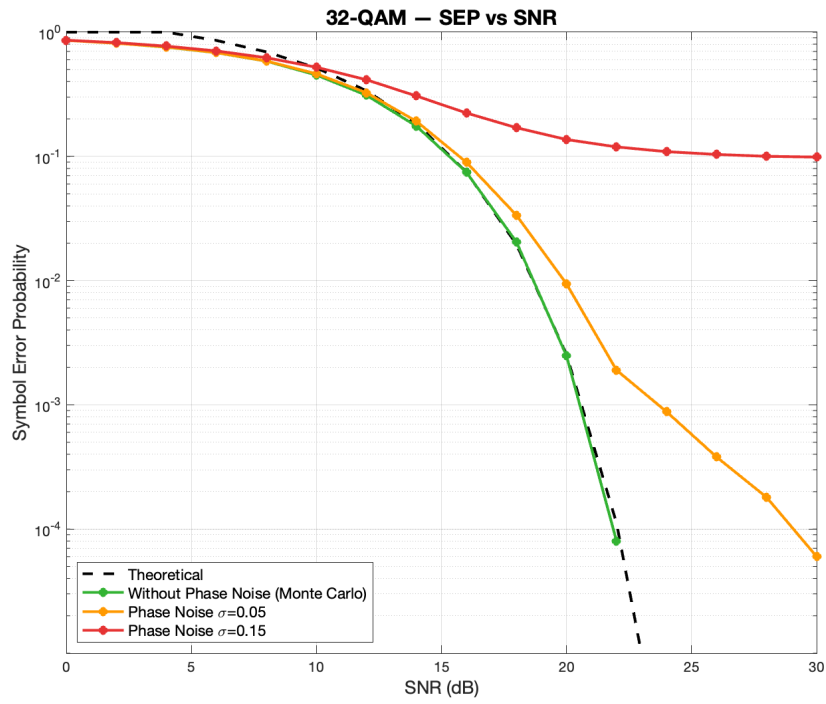


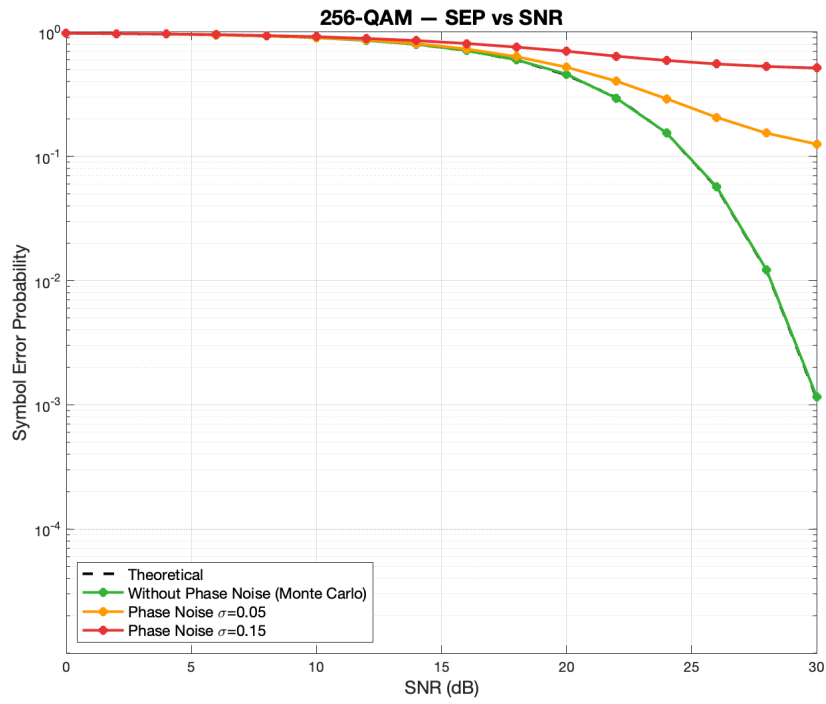
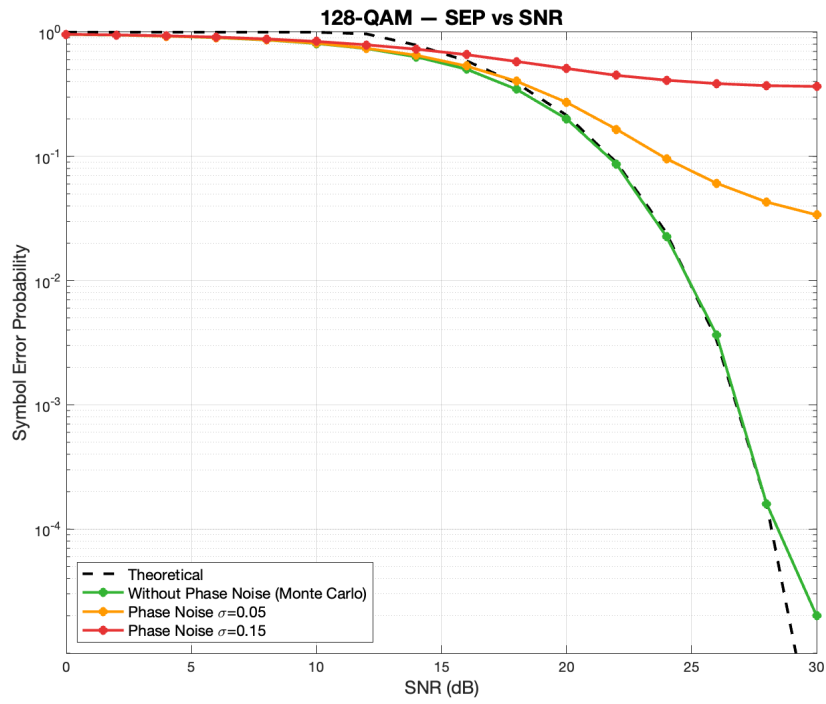
Figure 2: 128-APSK modulation, severe impairments, very high SNR

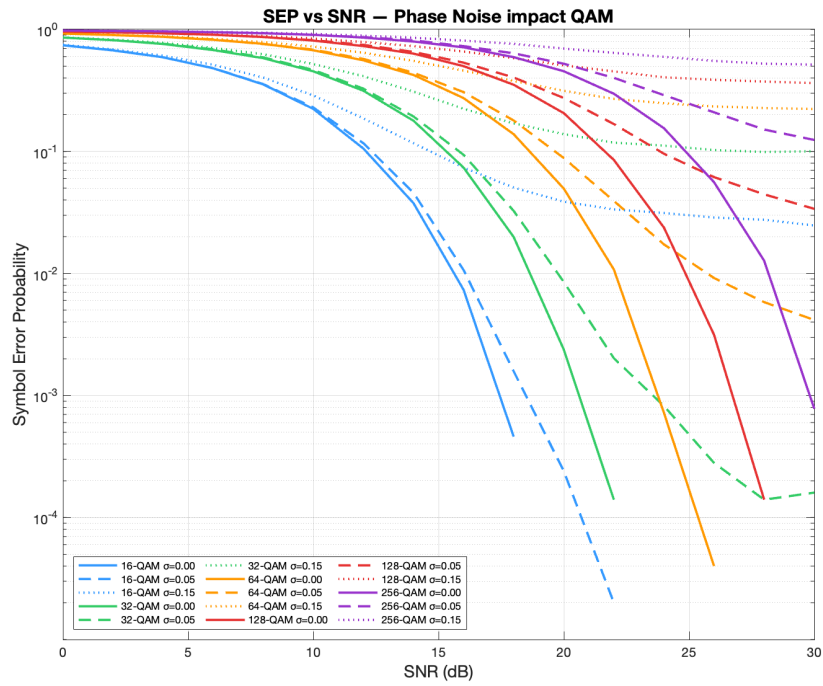
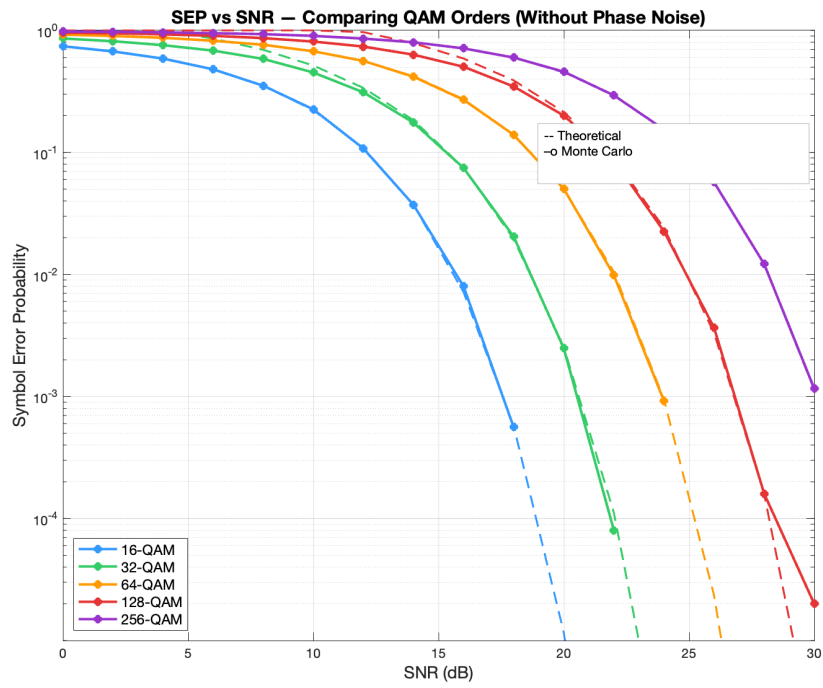
where $d_{\min}$ is the minimum Euclidean distance of the constellation and $\bar{N}$ is the average number of nearest neighbors. In the presence of phase noise, no closed-form expression exists for any order; Monte Carlo simulation with $N = 50,000$ symbols per SNR point was used exclusively.

The SEP curves confirm the expected behavior: higher-order constellations require significantly higher SNR to achieve the same error probability, and phase noise introduces an irreducible error floor that persists even at high SNR values. This floor is more pronounced for higher-order constellations, as their denser symbol packing makes them more sensitive to phase disturbances. The results are presented in the following Figures.









2 Part II: Model Training and Evaluation

2.1 Convolutional Neural Networks

2.1.1 Architectural Overview

Convolutional Neural Networks (CNNs) were introduced by Yann LeCun in the late 1980s [4]. They are a class of deep learning models specifically designed for processing grid-structured data such as images. Unlike fully connected networks, which treat each input pixel independently, CNNs exploit the spatial locality of visual patterns through three core operations: convolution, activation, and pooling.

A convolutional layer applies a set of learnable filters $\mathbf{W} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k \times k}$ to the input feature map $\mathbf{X}$, producing an output feature map through the discrete cross-correlation:

$$\mathbf{Y}[c, i, j] = \sum_{c'} \sum_{p=0}^{k-1} \sum_{q=0}^{k-1} \mathbf{W}[c, c', p, q] \cdot \mathbf{X}[c', i+p, j+q] + b_c \quad (17)$$

where k is the filter size, c and c' index the output and input channels respectively, and b_c is a learnable bias. Each filter learns to detect a specific local pattern, and the spatial sharing of filter weights across the entire image dramatically reduces the number of trainable parameters compared to a fully connected layer.

Following each convolutional layer, a non-linear activation function is applied element-wise to introduce the representational capacity necessary for modeling complex decision boundaries. The Rectified Linear Unit (ReLU) is the standard choice:

$$\text{ReLU}(x) = \max(0, x) \quad (18)$$

Pooling layers reduce the spatial dimensions of the feature maps, providing a degree of translation invariance and reducing the computational cost of subsequent layers. Max pooling retains the most prominent activation within each local window:

$$\mathbf{Y}[c, i, j] = \max_{0 \leq p, q < k} \mathbf{X}[c, si+p, sj+q] \quad (19)$$

where s is the stride. The hierarchical stacking of convolutional, activation, and pooling layers progressively transforms low-level pixel patterns into high-level semantic representations. The final layers of a CNN are typically fully connected, mapping the extracted feature vector to the output space.

2.1.2 Multi-Task Learning Formulation

In the standard single-task classification setting, a CNN maps an input image $\mathbf{x}$ to a single label y . In this work, each constellation image is associated with five distinct classification targets simultaneously:

$$\hat{\mathbf{y}} = (\hat{y}_{\text{mod}}, \hat{y}_{\text{pn}}, \hat{y}_{\text{iq}}, \hat{y}_{\text{jam}}, \hat{y}_{\text{snr}}) \quad (20)$$

corresponding to modulation type (16 classes), phase noise severity (4 classes), I/Q imbalance severity (4 classes), jamming severity (4 classes), and SNR range (5 classes) respectively. This formulation is known as *multi-task learning*, where a shared feature extractor (the convolutional backbone) feeds into multiple task-specific classification heads, one per label. The total training loss is the sum of the individual cross-entropy losses:

$$\mathcal{L}_{\text{total}} = \sum_{t \in \mathcal{T}} \mathcal{L}_{\text{CE}}^{(t)}, \quad \mathcal{L}_{\text{CE}}^{(t)} = - \sum_{c=1}^{C_t} y_c^{(t)} \log \hat{p}_c^{(t)} \quad (21)$$

where $\mathcal{T}$ is the set of tasks, C_t is the number of classes for task t , $y_c^{(t)}$ is the one-hot ground truth, and $\hat{p}_c^{(t)}$ is the predicted probability for class c .

2.1.3 Experimental Motivation

The choice of architectures to compare was motivated by the findings of [2], in which the authors evaluated LeNet (LN) and AlexNet (AN) for automatic modulation classification. Their results demonstrated that the lighter LeNet architecture consistently outperformed the significantly more complex AlexNet, leading the authors to conclude that “*simpler CNN architectures such as LN can achieve better accuracy than more complex ones such as AN, thus challenging the popular notion that more complex architectures inherently lead to better performance*” [2].

The present work extends this investigation to the multi-task learning setting, where the model is required to simultaneously predict five labels per constellation image. The central research question is whether the advantage of lightweight architectures observed in single-task modulation classification generalizes to the more demanding multi-task scenario. To this end, LeNet was compared against ResNet-18, a residual network that is substantially deeper and more parameterized, under identical training conditions.

2.2 Dataset Split

The full dataset of 32,768 constellation images was partitioned into training, validation, and test sets using a stratified split to ensure balanced class representation across all partitions. The split ratios were 70% training, 15% validation, and 15% testing, yielding approximately 22,937 training images, 4,915 validation images, and 4,916 test images. The split was performed at the image level, ensuring that all five question-answer pairs associated with a given image belong exclusively to one partition, thus preventing any form of data leakage between sets.

2.3 LeNet: Detailed Architectural Analysis and Multi-Task Implementation

2.3.1 Evolutionary Background and Architectural Foundations

The LeNet-5 architecture, pioneered by Yann LeCun in 1998 [4], represents a seminal milestone in the field of deep learning, establishing the fundamental template for modern convolutional neural networks (CNNs). Originally designed for the classification of handwritten digits in the MNIST dataset, LeNet introduced the concept of local receptive fields and shared weights to exploit the spatial correlations inherent in image data. In the context of this research, we utilize an evolved variant, hereafter referred to as LeNet, which adapts the classic convolutional backbone to the requirements of high-dimensional RF constellation analysis.

The choice of LeNet is strategically motivated by its computational efficiency and its ability to act as a baseline for spatial feature extraction. While modern architectures like ResNet or Transformers utilize hundreds of millions of parameters, LeNet relies on a shallower hierarchical structure, which allows for an unclouded observation of how basic convolutional kernels perceive signal impairments. The architectural layout, as depicted in the block diagram of Figure 3, follows a structured pipeline of feature transformation.

The shared feature extractor is composed of the following sequential stages:

- 1. Initial Feature Extraction (C1 Layer):** A convolutional layer utilizing 5×5 kernels. This stage is responsible for detecting primary edges and circular boundaries within the polar constellation plots.
- 2. Non-Linear Activation and Subsampling (S2 Layer):** The Rectified Linear Unit (ReLU) function is applied to introduce non-linearity, followed by a 2×2 Max-Pooling layer. This subsampling stage is critical as it reduces the sensitivity of the filters to the exact pixel location of the scatter points, providing initial translation invariance.
- 3. Higher-Order Feature Integration (C3 Layer):** A second convolutional layer that increases the depth of the feature maps, allowing the network to combine simple edge

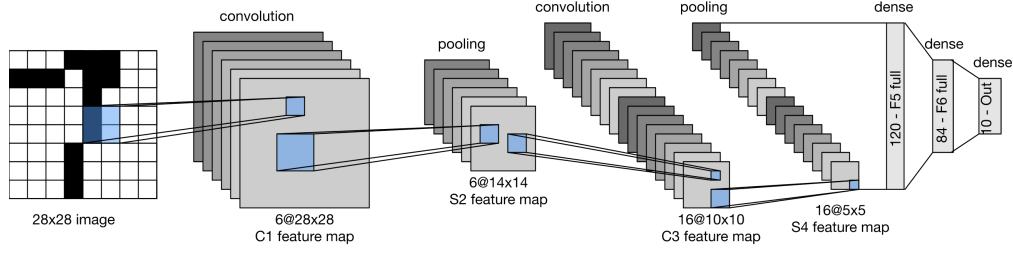


Figure 3: Comprehensive block diagram of the LeNet architecture. The diagram illustrates the transition from the input image through the shared convolutional stages to the multi-task classification heads.

detections into more complex geometric representations, such as the concentric rings of APSK or the grid-like patterns of QAM.

4. **Shared Fully Connected Layers (F4 and F5):** After flattening the spatial maps, two dense layers with 120 and 84 neurons respectively serve as the "associative memory" of the network, integrating the extracted features into a high-level representation before the task-specific branching.

2.3.2 Multi-Task Learning and Training Dynamics

The implementation was carried out using the PyTorch framework [5], employing an end-to-end optimization strategy. Unlike standard classification, our LeNet model is equipped with five distinct "output heads." Each head consists of a linear layer followed by a Softmax activation, independently calculating the probability distribution for Modulation, Phase Noise, I/Q Imbalance, Jamming, and SNR Range.

The training process spanned 15 epochs, utilizing a batch size of 32 images. The total loss function was optimized using the Adam optimizer, which adaptively adjusts the learning rate for each parameter. The training trajectory, as summarized in Table 4, shows a significant drop in loss within the first five epochs, indicating that the network quickly identified the dominant geometric features of the constellations.

The validation loss closely followed the training loss, suggesting that the model successfully generalized the learned patterns without falling into the trap of catastrophic overfitting. The small fluctuations in validation loss toward the final epochs (12-15) indicate that the model reached its representational capacity, settling into a stable local minimum.

2.3.3 Analysis of Test Performance and Error Modalities

Upon completion of the training phase, the LeNet model was subjected to a rigorous evaluation using the held-out test set. The resulting accuracies across the five classification tasks provide deep insights into the relationship between the CNN architecture and the physical properties of RF signals represented as spatial distributions:

- **Jamming Detection (89.16%):** The network achieved its highest accuracy in this task. Jamming typically manifests as a wide, diffused interference ring or a broad macroscopic alteration to the overall texture of the constellation. Because convolutional filters excel at identifying large-scale geometric patterns and global textures, LeNet successfully detects the presence and severity of jamming interference with high confidence.

Table 4: Detailed Progression of Training and Validation Loss (Updated Dataset)

Epoch	Training Loss	Validation Loss
1	7.2075	6.1389
2	5.7041	5.3834
3	5.1720	5.0359
4	4.8539	4.7396
5	4.6270	4.5793
6	4.4499	4.4912
7	4.3168	4.3564
8	4.2021	4.2998
9	4.1076	4.2133
10	4.0260	4.2033
11	3.9571	4.0546
12	3.8868	4.0790
13	3.8250	4.1173
14	3.7648	4.0076
15	3.7029	3.9635

- **SNR Range Estimation (80.04%):** The model demonstrates robust performance in estimating the Signal-to-Noise Ratio. SNR directly dictates the variance, spatial spread, and pixel density of the constellation's probability distribution. LeNet is highly capable of measuring this global dispersion, effectively evaluating how "tight" or "scattered" the overall noise cloud is across the spatial domain.
- **Modulation Classification (64.12%):** The moderate accuracy in this category reflects the inherent challenge of purely spatial demodulation. The network must infer the modulation order entirely from the overlapping clusters of the noise clouds. While it successfully distinguishes fundamentally different shapes (e.g., PSK vs. QAM), the overlapping probability distributions in higher-order schemes (such as distinguishing 64-QAM from 256-QAM under noise) make cluster separation increasingly difficult for standard spatial filters.
- **I/Q Imbalance (55.66%):** Performance degrades noticeably when evaluating I/Q Imbalance. This impairment causes an asymmetrical, non-uniform compression or stretching of the constellation along specific axes. Estimating the exact severity of this distortion relies purely on calculating the precise asymmetry of the overall noise blob, a task where shallow convolutional layers begin to lose resolution.
- **Phase Noise (33.93%):** The severe drop in accuracy highlights the most critical vulnerability of the architecture. Phase Noise introduces rotational smearing around the origin. Determining the exact angular deviation of a noise cloud without temporal or phase tracking is exceptionally challenging for a vision-based network.

The degraded performance in estimating Phase Noise and I/Q Imbalance highlights a fundamental limitation of the LeNet architecture: Pooling operations, such as Max-Pooling, are inherently designed to discard exact spatial offsets in order to make the network robust to translations. However, in RF diagnostics, the severity of transceiver impairments is defined by exact pixel-level rotational and compressive micro-transformations. By the time the signal representation reaches the fully connected heads, the pooling layers have "smeared" these fine-grained geometric offsets. This analysis concludes that while LeNet is highly effective at identifying global channel conditions (SNR, Jamming), purely spatial CNNs struggle to classify subtle, micro-spatial transceiver impairments due to their inherent architectural invariance.

2.3.4 In-depth Confusion Matrix Analysis and Feature Correlation

To obtain a comprehensive understanding of LeNet’s decision-making process, we examine the confusion matrices for each individual task. This visual analysis reveals how the network handles class boundaries and highlights the specific topological features that the convolutional kernels struggle to separate without ideal reference points.

Note: The categorical axes in the following matrices are plotted based on the model’s internal class indexing rather than sequential severity, requiring careful observation of the specific label intersections.

Modulation Identification As illustrated in Figure 4, the confusion matrix for modulation classification exhibits a heavily fragmented diagonal.

Confusion Matrix: Modulation

True Label \ Predicted Label	128-APSK	128-QAM	16-APSK	16-HQAM	16-QAM	256-QAM	32-APSK	32-QAM	4-ASK	4-HQAM	64-APSK	64-HQAM	64-QAM	8-ASK	BPSK	QPSK
128-APSK	120	28	27	12	7	1	30	21	0	1	43	5	8	0	0	5
128-QAM	14	151	19	10	5	10	7	46	0	0	16	1	7	0	0	9
16-APSK	6	17	188	4	3	0	15	16	0	0	17	8	9	0	0	5
16-HQAM	6	6	3	222	2	3	1	24	0	11	2	4	9	0	0	6
16-QAM	5	5	8	8	218	17	2	12	0	1	1	0	28	0	0	17
256-QAM	2	18	5	17	30	99	1	15	0	0	3	0	109	0	0	14
32-APSK	31	31	28	6	3	1	138	24	0	0	25	4	5	0	0	0
32-QAM	5	43	20	15	12	4	2	176	0	0	12	2	10	0	0	8
4-ASK	0	0	0	0	0	0	0	0	281	0	0	0	0	40	7	0
4-HQAM	1	0	0	17	0	0	0	0	0	296	0	1	1	0	0	1
64-APSK	95	25	27	7	2	1	30	15	0	0	107	5	2	0	0	3
64-HQAM	11	6	26	18	0	3	11	6	0	1	23	179	2	0	0	3
64-QAM	5	13	6	7	38	24	2	9	0	0	5	0	179	0	0	22
8-ASK	0	0	0	0	0	0	0	0	44	0	0	0	0	253	3	0
BPSK	0	0	0	0	0	0	0	0	21	0	0	0	0	3	267	0
QPSK	0	3	1	3	12	9	0	7	0	2	3	0	14	0	0	278

Figure 4: Confusion Matrix for Modulation Classification. Strong predictive accuracy is isolated to lower-order schemes, with severe inter-class leakage among dense high-order constellations.

The matrix clearly shows that the model successfully isolates lower-order modulations (such as BPSK, QPSK, and low-order ASK), where the geometric shape of the noise clouds remains distinct. However, the main confusion emerges among the higher-order dense schemes (e.g., 64-QAM, 128-QAM, 256-QAM, and the APSK families). The overlapping Gaussian distributions of these dense schemes look virtually identical to the network’s spatial filters, forcing it to guess between physically similar probability distributions.

External Interference: Jamming Severity Conversely, the detection of Jamming interference (Figure 5) presents a highly structured matrix, demonstrating the model’s proficiency in macroscopic texture recognition.

The network easily isolates the "Medium" and "Severe" states, achieving near-perfect true positive rates. Jamming introduces a global structural change to the entire image (often appearing as widespread interference rings). The hierarchical convolutional filters maintain a high response to these broad changes. The few misclassifications that do occur are strictly constrained to confusing "None" with "Mild", which is expected when transitioning through continuous interference power levels.

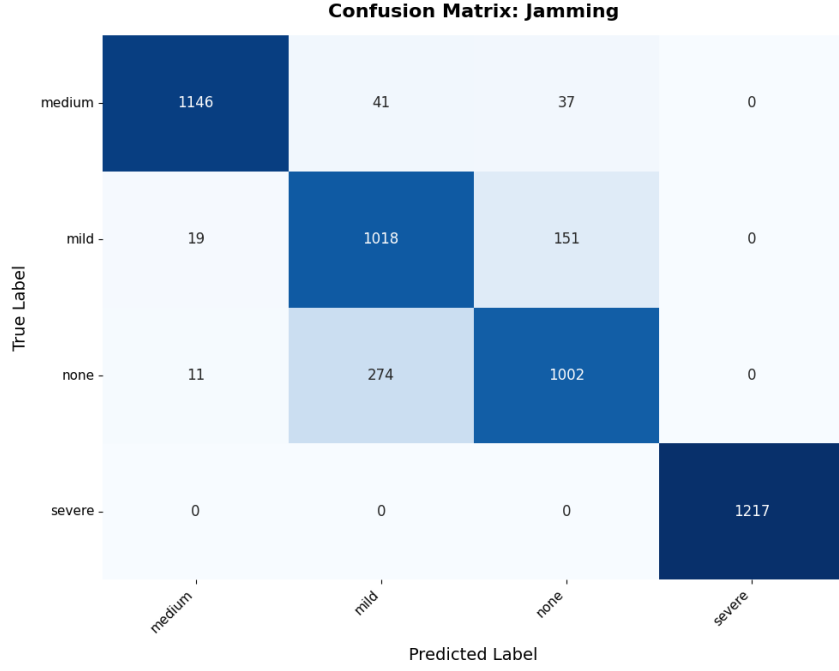


Figure 5: Confusion Matrix for Jamming Severity. The strong diagonal confirms the network’s ability to consistently track global interference patterns, with errors limited to adjacent classes.

SNR Range Estimation The SNR classification matrix (Figure 6) provides insight into how the network evaluates global spatial variance.

The matrix reveals a classic "adjacent class leakage" pattern. The network rarely suffers from catastrophic errors (e.g., confusing "Very Low" with "Very High" SNR). Instead, the ambiguity is concentrated directly on the borders between adjacent ranges. This indicates that LeNet successfully approximates the overall density and spread of the noise clouds, but acts as a coarse thresholding classifier rather than a precise statistical estimator of continuous noise variance.

Transceiver Impairments: The Micro-Spatial Breakdown The matrices for I/Q Imbalance (Figure 7) and Phase Noise (Figure 8) depict a severe breakdown in discriminatory power, highlighting the limitations of spatial invariance.

The I/Q Imbalance matrix shows a significantly degraded diagonal. Because I/Q Imbalance causes an asymmetrical, axis-dependent compression of the constellation, measuring it requires a defined geometric center. Without it, the network struggles to differentiate between an inherently asymmetrical noise cloud and actual transceiver distortion.

The Phase Noise matrix represents a near-catastrophic collapse. The distribution of predictions is heavily scattered and heavily biased toward the "None" class, indicating that the network is essentially guessing. Phase noise manifests as a continuous rotational variance. The convolutional and max-pooling layers of LeNet are designed to ignore slight spatial shifts. Consequently, the network entirely misses the sub-pixel rotational smearing, proving that traditional CNNs are fundamentally unsuited for blind phase estimation.

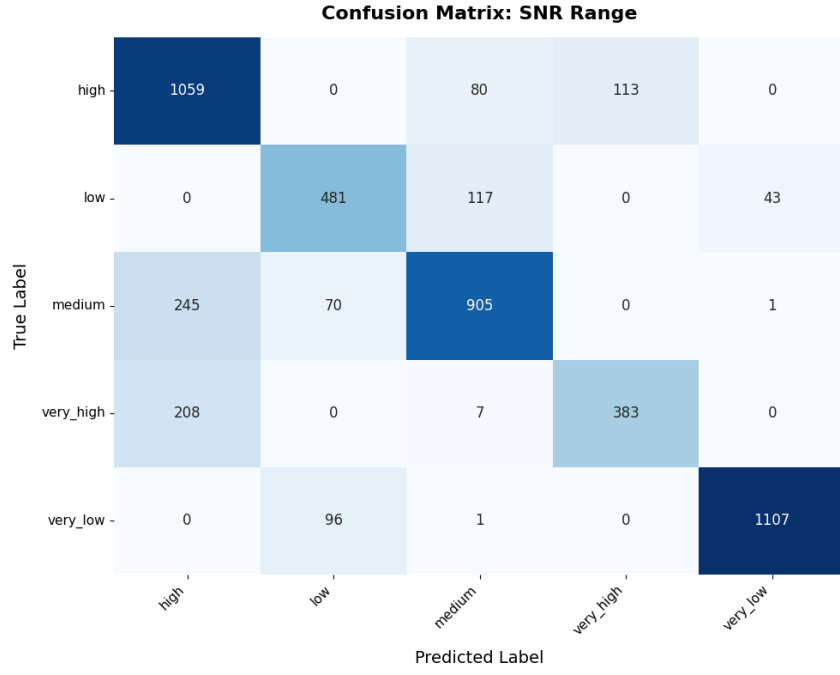


Figure 6: Confusion Matrix for SNR Range. The network demonstrates classic adjacent-class leakage, successfully capturing the overall spatial density but struggling at exact categorical thresholds.

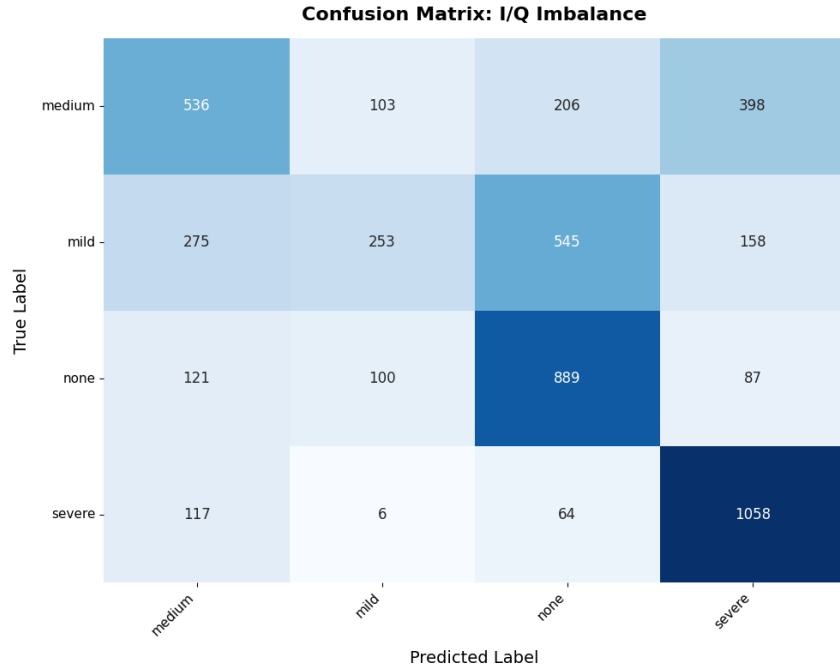


Figure 7: Confusion Matrix for I/Q Imbalance. The fading diagonal and high false-positive rates for baseline classes indicate a failure to quantify non-uniform spatial compression.

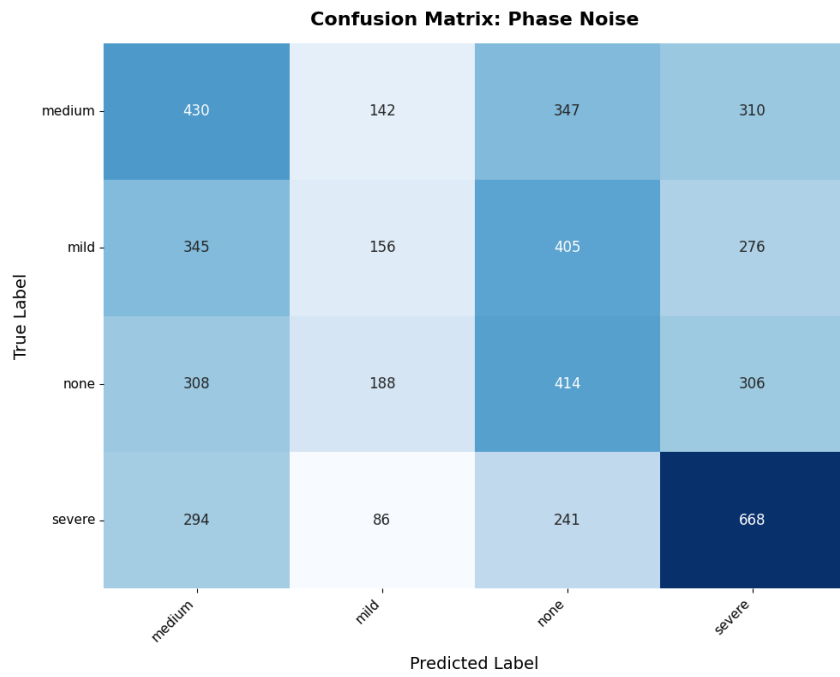


Figure 8: Confusion Matrix for Phase Noise. The matrix reflects a state of near-random guessing and severe bias, confirming the network’s inability to extract micro-rotational features blindly.

2.3.5 Generalization and Robustness Testing on Out-of-Distribution Data

To rigorously assess the true generalization capabilities of the LeNet architecture, an Out-Of-Distribution (OOD) robustness test was conducted. A dedicated test set of 400 novel images was generated using intermediate and slightly shifted parameter values that the network had not explicitly encountered. Specifically, the Signal-to-Noise Ratio (SNR) evaluated intermediate points such as 3, 7, 12, 18, 24, 29, 33, and 38 dB, testing the boundaries of the predefined SNR categories. Furthermore, transceiver impairments were shifted to new continuous states, including modified Phase Noise variances (ranging up to 0.13 radians), altered I/Q Imbalance parameters (gain imbalance up to 1.80 and phase imbalance up to 9 degrees), and varied Jamming powers (up to 9 dB).

This test determines whether the network learned the underlying physical transformations of the RF signals or simply overfit to specific, discrete visual footprints. The evaluation yielded the following insights:

- **Jamming Detection (89.00%):** The model demonstrated exceptional robustness in this task. Because jamming introduces a severe, macroscopic topological change (such as widespread circular interference rings), shifting the exact jamming power (e.g., to 1 or 9 dB) does not alter the fundamental visual signature. LeNet successfully generalized its understanding of this global texture.
- **SNR Range Estimation (69.00%):** The accuracy reflects a noticeable decline compared to standard testing but remains highly functional. The introduction of intermediate SNR values (e.g., 7 dB or 12 dB) forces the network to evaluate noise distributions that lie directly on or near the decision boundaries of predefined categories (e.g., distinguishing between 'low' and 'medium').
- **Modulation Classification (56.25%):** A drop in modulation accuracy is demonstrated. By combining intermediate SNR levels with shifted distortion parameters, the visual geometry of the constellation clouds morphs into atypical shapes. The spatial filters of the CNN, having memorized specific discrete visual templates, struggle to classify these newly formed, overlapping probability distributions accurately.
- **I/Q Imbalance (55.25%):** : The network's performance on this task remains practically equal to the initial testing. The same difficulties that were discussed in the previous testing apply, making it hard for the model to successfully predict the correct labels for I/Q imbalance.
- **Phase Noise (29.75%):** : The performance features a small drop compared to the already low 33.39% in the initial testing. The OOD pictures prove to be even more difficult for the model to generalize and thus the performance plummets even further.

In conclusion, the OOD robustness test reveals that the model's performance remains pretty much the same when compared to the initial testing, with the 2 exceptions being Modulation and SNR range classification. This indicates robustness and the model's ability to correctly generalize even under conditions not seen during training.

2.4 ResNet-18: Residual Learning and Regularization Strategies

2.4.1 Architectural Innovation: Deep Residual Learning

The ResNet (Residual Network) architecture, introduced by He et al. in 2016 [6], represented a paradigm shift in deep convolutional networks by addressing the vanishing gradient problem in very deep architectures. Unlike traditional "plain" networks such as LeNet, ResNet-18 utilizes "Residual Blocks" based on skip connections or identity mappings.

The core principle behind this innovation is the ability of the network to learn a residual mapping $\mathcal{F}(\mathbf{x})$ rather than the full underlying mapping $\mathcal{H}(\mathbf{x})$. Formally, the output is defined as:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \quad (22)$$

ResNet-18 consists of 18 weighted layers, including eight residual blocks that allow the gradient to flow through the network more effectively during backpropagation. For this multi-task study, the standard ResNet-18 backbone was modified by replacing the final fully connected layer with five task-specific heads, mirroring the multi-task configuration used in the LeNet experiments.

2.4.2 Combatting Overfitting through Regularization

Due to the significantly higher parameter count and representational capacity of ResNet-18 compared to LeNet, initial training sessions exhibited pronounced signs of overfitting. While the training loss decreased rapidly, the validation loss began to diverge, indicating that the model was memorizing specific samples rather than learning generalized RF features.

To mitigate this, a second, optimized training session was conducted over 15 epochs (approximate duration: 25 minutes) utilizing several regularization techniques:

- **Weight Decay (L_2 Regularization):** A penalty term was added to the loss function proportional to the square of the weights. This encourages smaller weight values, effectively limiting the complexity of the model's decision boundaries.
- **Dropout:** Layers were added that randomly deactivate a fraction of neurons ($p = 0.5$) during each training pass. This prevents "co-adaptation" of features, forcing the network to learn more robust and redundant representations.
- **Model Selection (Checkpointing):** To ensure optimal generalization, the model parameters were monitored throughout the 15 epochs, and the state corresponding to the minimum loss point was retained for final evaluation.

2.4.3 Training Progression and Convergence Analysis

The training trajectory of the optimized ResNet-18 is detailed in Table 5. The introduction of Dropout and Weight Decay resulted in a much more controlled convergence.

The data shows that optimal validation performance was achieved around Epoch 6 ($L_{val} = 2.9304$), after which the training loss continued to fall while the validation loss began a steady climb. This behavior confirms the necessity of the regularization strategies, as the deep architecture reaches its peak generalization early before beginning to overfit the training distribution.

2.4.4 Evaluation Results

The evaluation of the ResNet-18 model on the test set yielded the following performance metrics across the five multi-task objectives. These results illuminate how deeper hierarchical feature extraction handles complex RF signal topologies, especially when it comes to Multi-Task 1:

- **Jamming Detection (92.33%):** The model achieves excellent accuracy in identifying jamming. This interference creates prominent, large-scale structural patterns (such as wide circular halos) across the constellation. The deep convolutional layers of ResNet-18 are highly proficient at recognizing and isolating these macroscopic, global texture changes.

Table 5: Training and Validation Loss: Optimized ResNet-18 (Dropout + Weight Decay)

Epoch	Training Loss	Validation Loss
1	5.8014	4.1013
2	4.0731	3.4083
3	3.4620	3.1738
4	3.1270	2.9875
5	2.8687	3.1347
6	2.5973	2.9304
7	2.2884	3.2278
8	1.9971	3.1513
9	1.6944	3.6300
10	1.4231	3.7238
11	1.2148	3.9830
12	1.0589	3.9840
13	0.9477	4.4166
14	0.8098	4.6195
15	0.7422	4.8943

- **SNR Range Estimation (88.22%)**: ResNet-18 demonstrates a strong capability in quantifying spatial variance and noise density. The deep residual pathways allow the network to effectively measure the overall spread and entropy of the probability distributions, enabling highly accurate classification of the signal-to-noise boundaries.
- **I/Q Imbalance Accuracy (73.68%)**: The network shows a robust capacity to detect I/Q Imbalance. This impairment introduces an asymmetrical, axis-dependent compression to the signal clouds. The increased depth of ResNet-18, combined with its ability to preserve fine-grained gradients through residual connections, equips the model to capture this non-uniform geometric stretching effectively.
- **Modulation Accuracy (71.11%)**: The model achieves commendable accuracy in distinguishing modulation schemes based purely on spatial representations. While overlapping noise clouds in dense, high-order schemes present an inherent challenge, the deep hierarchical feature maps successfully extract complex cluster shapes and grid densities, allowing reliable classification for the majority of modulation formats.
- **Phase Noise Accuracy (45.52%)**: Although Phase Noise remains the most challenging task, the network’s performance is significantly above random guessing. Phase noise manifests as sub-pixel rotational smearing. While spatial pooling operations inherently discard exact angular offsets, the residual blocks of ResNet-18 manage to retain enough micro-rotational features to extract a partial, yet meaningful, understanding of the phase variance.

Overall, these metrics indicate that the architectural depth and residual connections of ResNet-18 provide a highly robust framework for multi-task RF classification. The network excels at global spatial analysis (Jamming, SNR) and demonstrates a sophisticated ability to disentangle overlapping geometric impairments (such as I/Q Imbalance) that typically confound shallower architectures.

2.4.5 Confusion Matrix Analysis and Feature Disentanglement

To evaluate the predictive reliability and decision-making process of the ResNet-18 architecture in a strictly blind classification scenario, we examine the confusion matrices for the five objec-

tives. These matrices offer a quantitative perspective on how deep residual features and identity mappings contribute to class separation when ideal spatial anchors are absent.

Note: The categorical axes in the following matrices are plotted based on the model’s internal class indexing rather than sequential severity, requiring careful observation of the specific label intersections.

Modulation Identification The confusion matrix for Modulation Classification (Figure 9) demonstrates a highly structured, albeit fragmented, diagonal corresponding to the 71.11% accuracy.

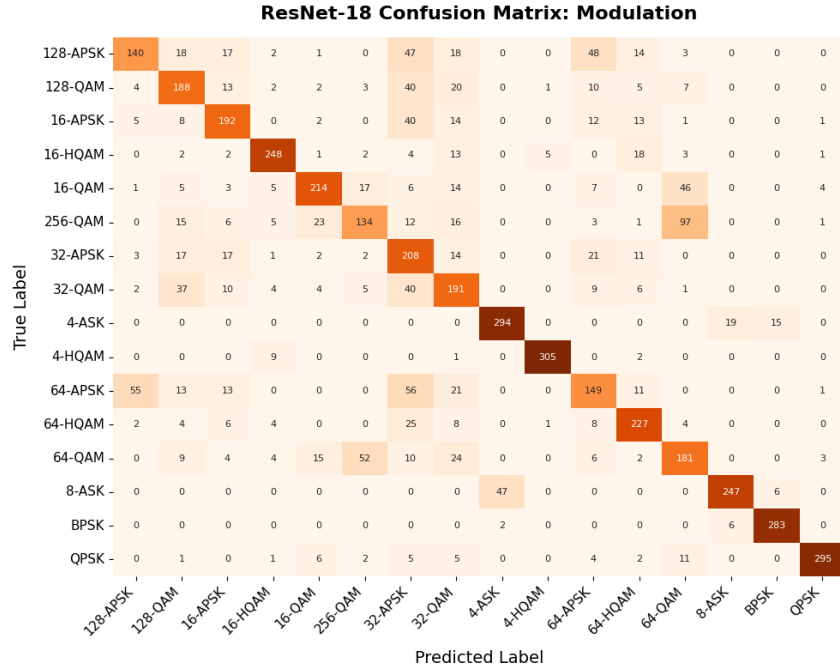


Figure 9: Confusion Matrix for ResNet-18 Modulation Classification. Deep residual features successfully isolate distinct geometric shapes, though confusion remains among dense, high-order alphabets.

The residual backbone proves highly effective at isolating lower-order and fundamentally distinct modulation formats. The deep hierarchical filters successfully extract complex cluster boundaries, allowing the network to recognize the topological differences between families (e.g., ASK vs. PSK). However, the matrix reveals distinct blocks of confusion among the highest-order schemes. The probability distributions of these dense constellations overlap significantly under noise, forcing the spatial filters into probabilistic guessing among physically similar visual clouds.

External Interference: Jamming Severity The classification of Jamming severity (Figure 10) shows exceptional predictive isolation.

Achieving a 92.33% accuracy, the true positive rates (the diagonal intersections for each specific label) are intensely concentrated. The deep convolutional layers are highly proficient at aggregating global spatial information, allowing them to easily identify the widespread circular interference rings indicative of jamming. Misclassifications are exceedingly rare and primarily occur between closely related interference thresholds, confirming the architecture’s robustness to macroscopic topological changes.

Precise SNR Range Estimation As illustrated in Figure 11, the SNR Range estimation exhibits strong categorical alignment.

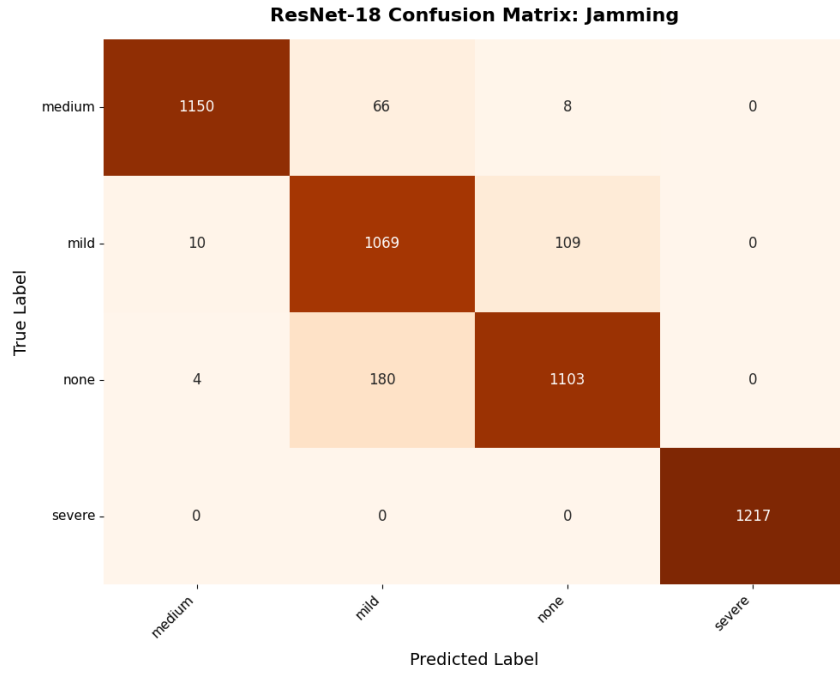


Figure 10: Confusion Matrix for ResNet-18 Jamming Severity. The network exhibits sharp categorical isolation, accurately detecting global interference halos regardless of axis plotting order.

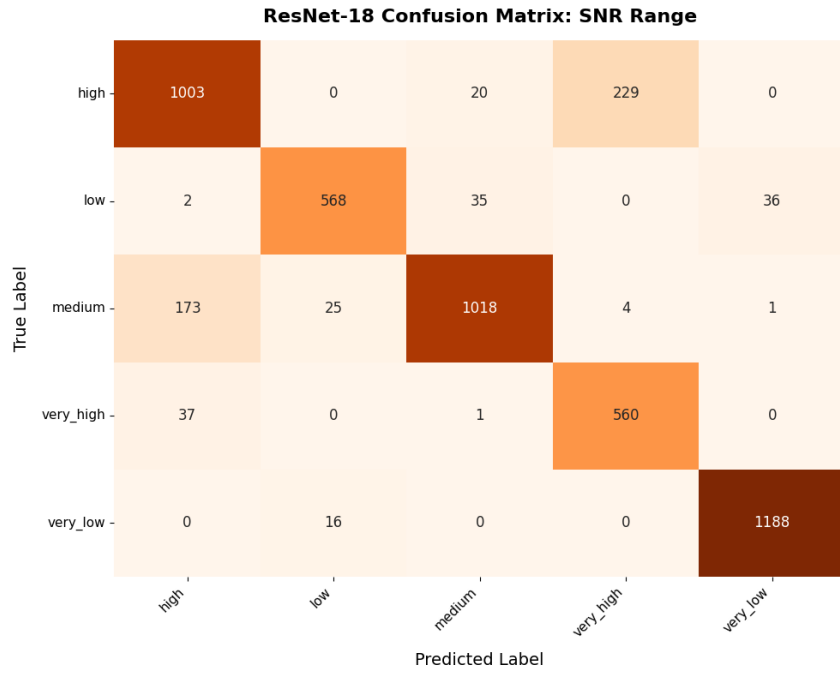


Figure 11: Confusion Matrix for ResNet-18 SNR Range. The deep feature hierarchy effectively quantifies spatial variance, maintaining high true-positive rates across all noise density thresholds.

The 88.22% accuracy reflects the deep network’s ability to act as a sophisticated statistical estimator. By preserving fine-grained gradients through its residual shortcuts, ResNet-18 accurately measures the overall entropy, point density, and spatial dispersion of the constellation clouds. The leakage outside the target intersections is minimal, indicating that the network has learned a highly discriminative latent representation of continuous noise variance.

I/Q Imbalance Characterization The matrix for I/Q Imbalance (Figure 12) highlights the advantage of deep feature extraction for axis-dependent distortions.

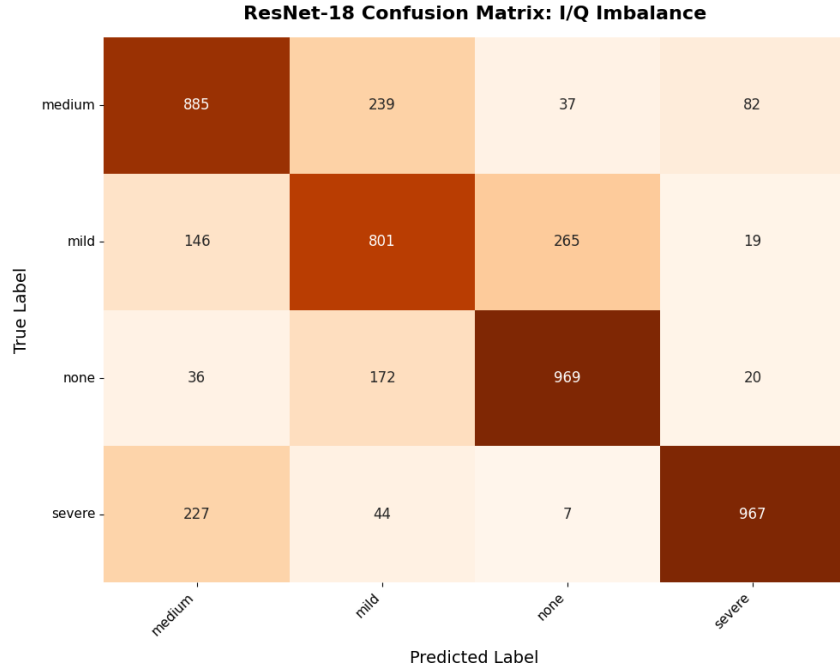


Figure 12: Confusion Matrix for ResNet-18 I/Q Imbalance. A strong diagonal structure confirms the network’s ability to detect non-uniform geometric compression using deep spatial gradients.

With an accuracy of 73.68%, the intersections for the specific severity labels show a solid predictive structure. I/Q Imbalance causes an asymmetrical, elliptical stretching of the signal. The increased depth of ResNet-18 allows it to maintain higher-resolution feature maps deep into the network, equipping the model to detect these non-uniform compressions even without a central point of reference.

Phase Noise Estimation Finally, the Phase Noise matrix (Figure 13) reflects the network’s performance on the most challenging impairment.

Although the overall accuracy is 45.52%, the true-positive intersections are visibly emerging from the background noise. Phase noise acts as a continuous rotational jitter. While spatial pooling operations inherently destroy exact angular coordinates, the identity mappings (skip connections) in ResNet-18 allow some critical micro-rotational features to bypass aggressive subsampling. Consequently, the network manages to extract a partial understanding of phase variance, though the notable dispersion confirms that purely visual classification of rotational smearing remains exceptionally difficult.

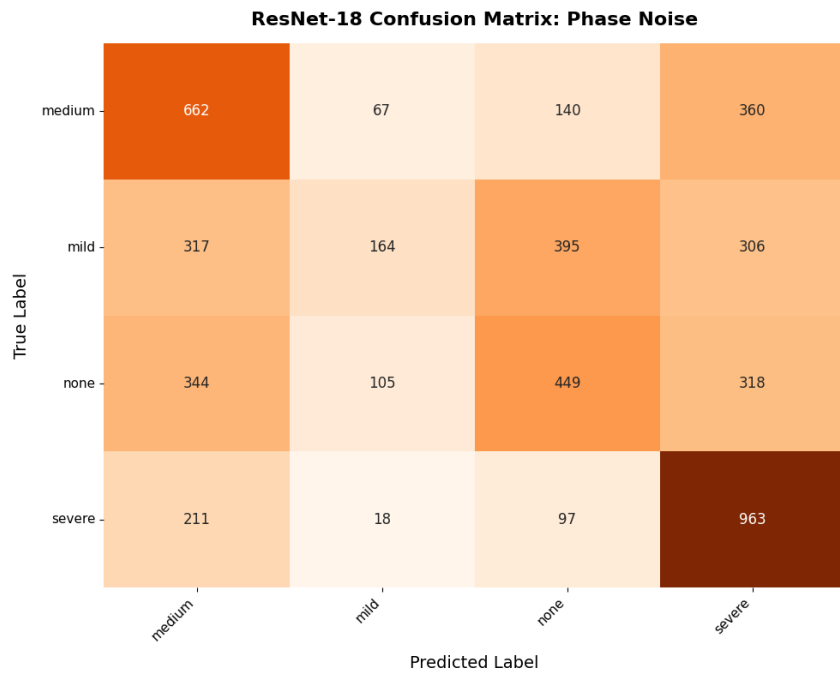


Figure 13: Confusion Matrix for ResNet-18 Phase Noise. While target intersections are identifiable, the high dispersion highlights the inherent difficulty of classifying sub-pixel rotational smearing without spatial anchors.

2.4.6 Generalization and Robustness Testing: Comparative Analysis

To rigorously evaluate the true robustness and generalization capacity of the deep residual architecture, the performance of the ResNet-18 model on the standard in-distribution test set was directly compared against its performance on the Out-Of-Distribution (OOD) test set. The OOD dataset introduces intermediate, continuous parameter shifts—such as unseen SNR tiers and novel transceiver distortion ratios—that force the network to operate outside its explicitly trained discrete boundaries.

By analyzing the performance shifts across each classification objective, we can determine how effectively the residual feature hierarchy generalizes to continuous RF environments:

- **Modulation Classification (71.50%):** The network demonstrates absolute resilience in this task. The statistical equivalence between the baseline and OOD datasets proves that ResNet-18 has not overfit to specific visual noise templates. Instead, the deep residual layers have learned the fundamental, generalized topological signatures of the constellation alphabets, allowing the model to accurately classify the modulation schemes regardless of the intermediate noise or interference boundaries.
- **Jamming Detection (91.50%):** Performance remains exceptionally stable. Because jamming introduces macroscopic, global interference halos to the image, the network easily tracks this overarching texture. Shifting the exact jamming power to unseen continuous values (e.g., intermediate dB levels) does not alter the fundamental structural signature of the interference, allowing the deep convolutional filters to generalize with near-perfect reliability.
- **Phase Noise Estimation (44.50%):** While the overall accuracy remains the lowest among the tasks, the stability across the two test sets is a highly positive architectural indicator. The network extracts whatever micro-rotational features it can preserve through its identity mappings. The fact that the accuracy does not degrade further in the OOD test confirms that the model’s partial understanding of phase variance is generalized, rather than memorized from discrete variance thresholds.
- **SNR Range Estimation (79.25%):** A noticeable degradation occurs in SNR classification. This drop is mathematically expected and highlights the inherent limits of ordinal classification on continuous spectra. The OOD test introduces intermediate SNR values (e.g., 7 dB or 12 dB) that land exactly on the ambiguous decision boundaries between pre-defined categories (such as ‘low’ and ‘medium’). While the network successfully measures spatial variance, forcing a rigid threshold on a continuous noise shift naturally increases classification ambiguity.
- **I/Q Imbalance Characterization (65.25%):** The network experiences a moderate decline in characterizing axis-dependent distortions. The OOD dataset introduces unseen combinations of gain and phase errors, generating non-standard elliptical compressions. Although ResNet-18 successfully detects the general presence of asymmetry, interpolating the exact severity of these novel geometric stretching ratios proves challenging, leading to boundary confusion between adjacent severity classes.

Ultimately, this comparative analysis highlights the structural robustness of architectural depth in RF classification. The deep identity mappings of ResNet-18 effectively mitigate the brittle generalization typically associated with vision-based signal analysis. While the model exhibits expected boundary confusion when thresholding continuous variables (SNR, I/Q Imbalance), it maintains unwavering robustness in identifying foundational topological structures (Modulation, Jamming).

2.5 Comparative Analysis: LeNet vs. ResNet-18

In this subsection, we conduct a comparison between the lightweight LeNet and the significantly deeper ResNet-18 architecture. This analysis evaluates the trade-offs between model complexity, computational requirements, and the ability to extract nuanced RF features under adverse channel conditions.

2.5.1 Overall Performance and Feature Resolution

The quantitative results on the test set are summarized in Table 6. The performance gap highlights the impact of architectural depth and residual learning on impairment classification.

Table 6: Comparative Accuracy Results on the Test Set

Objective	LeNet	ResNet-18	Improvement (Δ)
Modulation Type	64.12%	71.11%	+6.99%
Jamming Severity	89.16%	92.33%	+3.17%
SNR Range	80.04%	88.22%	+8.18%
I/Q Imbalance	55.66%	73.68%	+18.02%
Phase Noise	33.93%	45.52%	+11.59%

The comparative analysis reveals that the architectural depth of ResNet-18 provides the most substantial benefits when characterizing micro-spatial transceiver distortions. The most striking observation is the 18.02% improvement in I/Q Imbalance and the 11.59% increase in Phase Noise accuracy. While LeNet’s shallow architecture and aggressive spatial downsampling create a performance ceiling for detecting fine-grained geometric changes, the residual connections in ResNet-18 allow for a much more precise preservation of elliptical compression and rotational smearing. However, despite these notable improvements, Phase Noise remains the most challenging task for both architectures (peaking at 45.52%), confirming that sub-pixel rotational jitter is intrinsically difficult to differentiate from generalized Gaussian spread using purely visual, convolutional feature extractors.

2.5.2 Robustness across Noise Levels (Accuracy vs. SNR)

The robustness of each architecture was evaluated across various SNR levels. The following figures illustrate how each model maintains its predictive power as the channel conditions degrade.

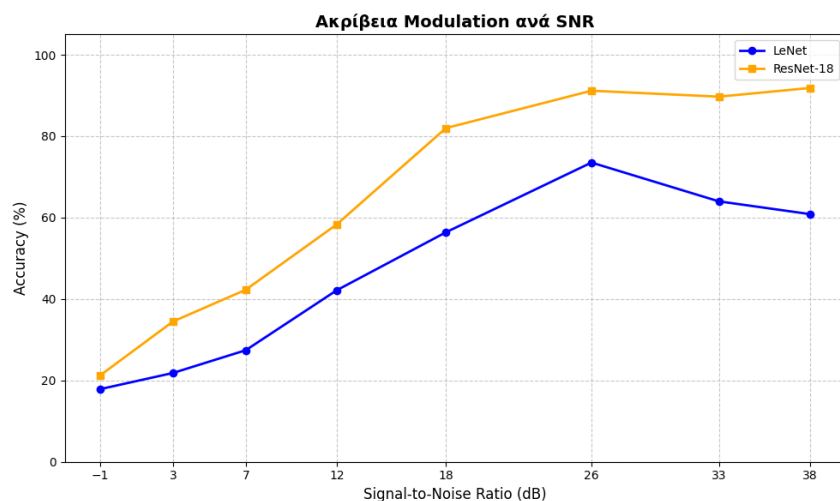


Figure 14: Accuracy vs. SNR: Modulation Classification.

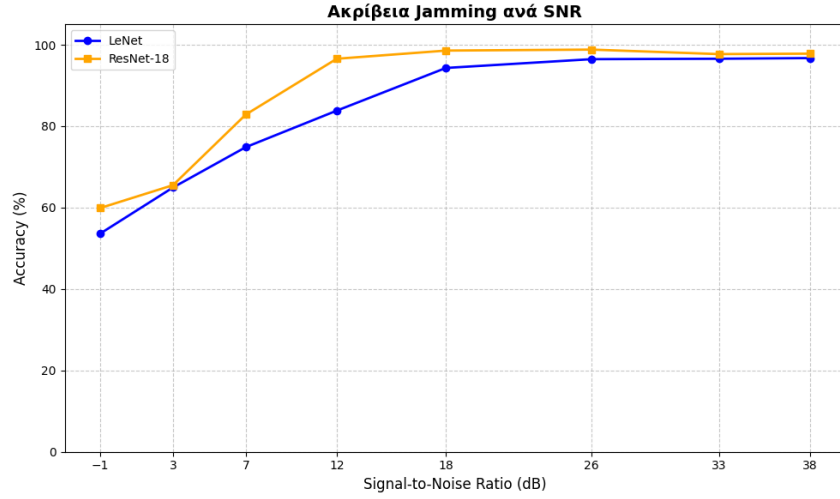


Figure 15: Accuracy vs. SNR: Jamming Detection. Both models are close, but ResNet-18 maintains a consistent lead, identifying interference patterns more effectively.

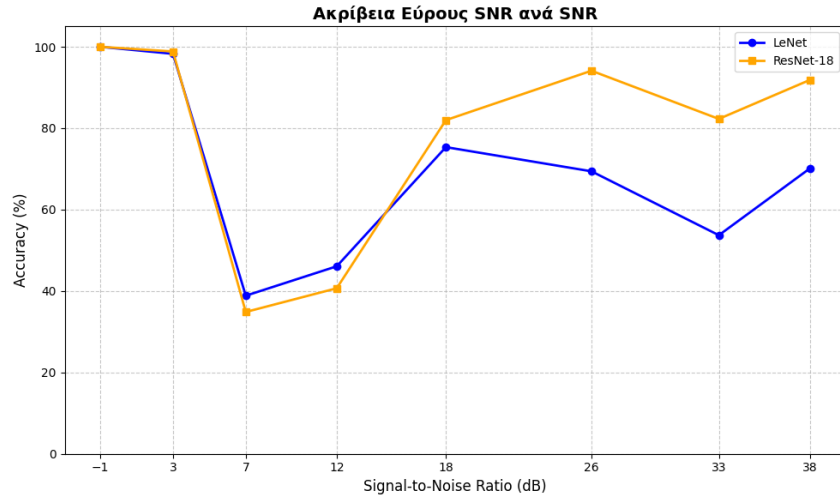


Figure 16: Accuracy vs. SNR: I/Q Imbalance. The significant performance gap illustrates that deeper architectures are essential for resolving axis-dependent distortions.

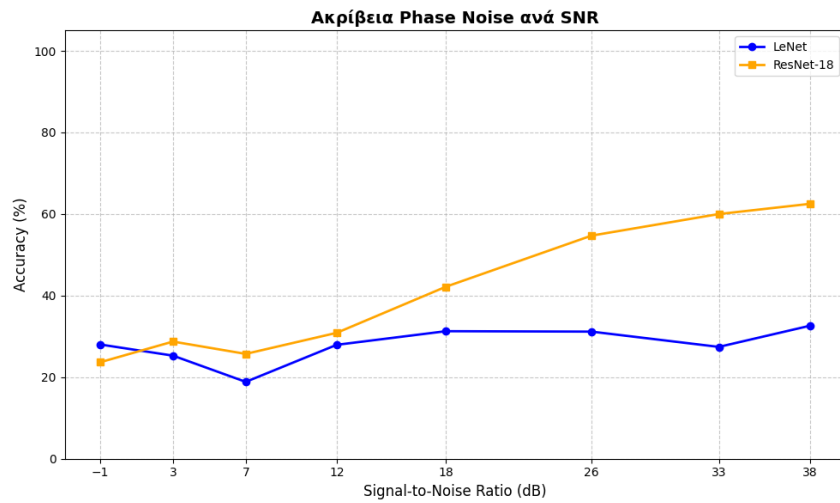


Figure 17: Accuracy vs. SNR: Phase Noise.

2.5.3 Computational Complexity and Deployment Considerations

The superior accuracy of ResNet-18 is balanced by its significantly higher computational overhead. The trade-off between the two models can be summarized as follows:

- **Parameter Density:** LeNet is a lightweight model with approximately 60,000 parameters, making it suitable for edge-deployment. ResNet-18, with roughly 11.7 million parameters, represents a $\sim 200\times$ increase in memory footprint.
- **Training Effort:** ResNet-18 required a more complex training strategy involving Dropout and Weight Decay to prevent overfitting, with training sessions lasting nearly three times as long as those for LeNet.
- **Inference Latency:** For real-time applications on software-defined radios (SDRs), LeNet offers lower latency. However, for post-processing and forensic signal analysis, ResNet-18 is the preferred choice due to its enhanced ability to disentangle overlapping impairments.

In conclusion, while LeNet is a highly efficient modulation classifier, ResNet-18 is necessary for higher-fidelity tasks requiring impairment estimation. The residual architecture effectively bridges the gap between purely visual pattern recognition and sub-pixel signal analysis.

2.6 Vision-Language Model Fine-Tuning

2.6.1 Vision-Language Models and Vision Transformers

Vision-Language Models (VLMs) are a class of multimodal neural architectures capable of jointly processing visual and textual inputs to produce natural language outputs. A VLM typically consists of three components: a vision encoder that extracts a sequence of visual features from an input image, a language model that processes text tokens, and a projection module that bridges the two modalities by mapping visual features into the token embedding space of the language model. In recent research, including the work known as RF-GPT [7], vision-language models have been adapted to interpret radio signal representations and answer structured questions about them. This raises the question, whether or not they can provide better performance than traditional Convolutional Neural Networks.

The vision encoder in modern VLMs is almost universally based on the Vision Transformer (ViT) architecture [8]. Unlike CNNs, which process images through local convolutional filters, a ViT divides the input image $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ into a sequence of non-overlapping patches of size $P \times P$ pixels:

$$N_p = \frac{H \cdot W}{P^2} \quad (23)$$

Each patch is flattened and linearly projected into a D -dimensional embedding vector. A learnable classification token [CLS] is prepended to the sequence, and positional embeddings are added to retain spatial information:

$$\mathbf{z}_0 = [\mathbf{x}_{\text{cls}}; \mathbf{E}\mathbf{x}_p^1; \mathbf{E}\mathbf{x}_p^2; \dots; \mathbf{E}\mathbf{x}_p^{N_p}] + \mathbf{E}_{\text{pos}} \quad (24)$$

where $\mathbf{E} \in \mathbb{R}^{D \times (P^2 C)}$ is the patch projection matrix. The resulting sequence is then processed by a stack of Transformer encoder blocks, each consisting of multi-head self-attention (MHSA) and a feed-forward network (FFN) with residual connections and layer normalization:

$$\mathbf{z}'_\ell = \text{MHSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1} \quad (25)$$

$$\mathbf{z}_\ell = \text{FFN}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell \quad (26)$$

The self-attention mechanism computes pairwise interactions between all patch tokens simultaneously, enabling the model to capture long-range spatial dependencies that are inaccessible to local convolutional filters:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (27)$$

[9]

where $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ are the query, key, and value matrices derived from linear projections of the input sequence, and d_k is the key dimension. A block diagram of the VLM architecture is illustrated in Figure 18.

2.6.2 SmolVLM-256M Architecture and Model Selection

SmolVLM-256M-Instruct [10] is a lightweight, open-source vision-language model developed by Hugging Face, comprising approximately 256 million parameters in total. Its architecture follows the standard VLM paradigm: a SigLIP-based vision encoder processes the input image, a linear projection maps visual tokens into the language model embedding space, and a SmolLM2-based causal language model generates the textual response autoregressively. The model was pretrained on large-scale image-text datasets and subsequently instruction-tuned to follow natural language

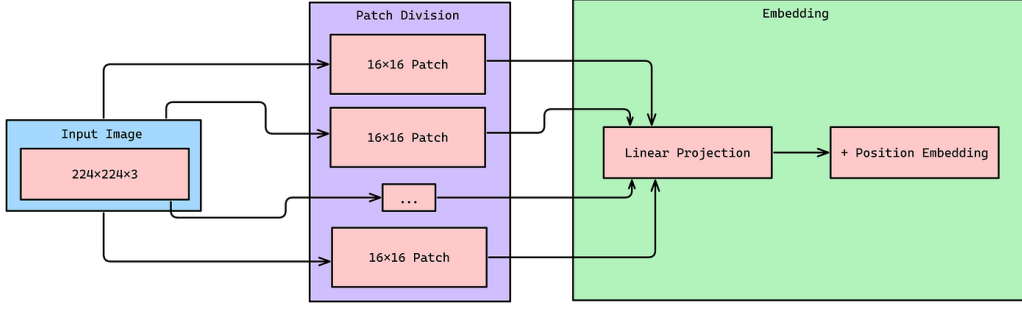


Figure 18: Block diagram of the Vision-Language Model architecture. The vision encoder (ViT) processes the constellation image into patch embeddings, which are projected into the language model token space and concatenated with the text query tokens before being processed by the autoregressive language model.

commands, making it directly applicable to visual question answering tasks without architectural modification.

The vision encoder divides input images into patches of 32×32 pixels. For a 256×256 input image, this yields a grid of $8 \times 8 = 64$ patch tokens, each compressed into a single embedding vector. The language model operates on a combined sequence of visual patch tokens and text tokens, generating the response one token at a time in an autoregressive fashion:

$$P(y_1, \dots, y_T | \mathbf{v}, \mathbf{q}) = \prod_{t=1}^T P(y_t | \mathbf{v}, \mathbf{q}, y_1, \dots, y_{t-1}) \quad (28)$$

where $\mathbf{v}$ denotes the sequence of visual tokens and $\mathbf{q}$ the tokenized text query. A block diagram of the SmolVLM-256M architecture is shown in Figure 19.

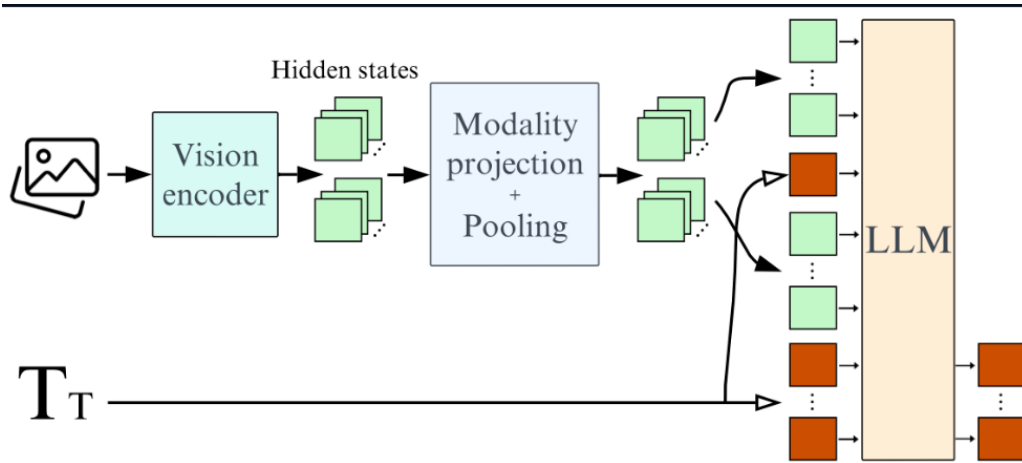


Figure 19: Block diagram of the SmolVLM-256M-Instruct architecture, showing the SigLIP vision encoder, the linear projection layer, and the SmolLM2 causal language model decoder.

SmolVLM-256M was selected for this work on the basis of three criteria. First, its parameter count of 256M makes it trainable on a single consumer GPU within a reasonable time budget, unlike larger VLMs which require multi-GPU infrastructure. Second, it is publicly available on the Hugging Face model hub and natively supports image-text instruction following, requiring no architectural modifications for the visual question answering formulation adopted here. Third, and most importantly, its relatively compact size makes it directly comparable to the CNN baselines in terms of the practical deployment constraints relevant to real-time RF signal processing systems.

2.6.3 Task Formulation

Rather than predicting a structured label vector as in the CNN multi-task setting, the VLM was formulated as a visual question answering system. Each constellation image was paired with five natural language questions, one per classification target, and the model was trained to produce a single-word answer for each:

- “What is the modulation scheme of this constellation?”
- “Identify the level of phase noise in this signal.”
- “What is the severity of the I/Q imbalance?”
- “Is there jamming present in this constellation? Describe it.”
- “Classify the Signal-to-Noise Ratio (SNR) range of this signal.”

Each (image, question, answer) triplet constitutes one training sample, so a single constellation image generates five training samples. The model was prompted with the system instruction: “You are an RF expert. Answer with one word only. No explanation.”, enforcing concise, structured output compatible with exact-match evaluation.

2.6.4 Parameter-Efficient Fine-Tuning via LoRA

Full fine-tuning of a 256M-parameter model is computationally prohibitive for the hardware budget available in this work. Low-Rank Adaptation (LoRA) [11] was adopted as the parameter-efficient fine-tuning strategy. LoRA freezes all pretrained weights and injects trainable low-rank decomposition matrices into the linear projection layers of the model. For a pretrained weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d \times k}$, the modified forward pass is:

$$\mathbf{W} = \mathbf{W}_0 + \Delta\mathbf{W} = \mathbf{W}_0 + \mathbf{B}\mathbf{A} \quad (29)$$

where $\mathbf{B} \in \mathbb{R}^{d \times r}$ and $\mathbf{A} \in \mathbb{R}^{r \times k}$ are the trainable LoRA matrices, with rank $r \ll \min(d, k)$. During training, $\mathbf{W}_0$ is kept frozen and only $\mathbf{A}$ and $\mathbf{B}$ are updated. The effective weight update is scaled by a factor α/r :

$$h = \mathbf{W}_0 x + \frac{\alpha}{r} \mathbf{B} \mathbf{A} x \quad (30)$$

The rank $r = 16$ and scaling factor $\alpha = 32$ were selected based on established practice in the VLM fine-tuning literature [11]. The rank $r = 16$ provides sufficient representational capacity for domain adaptation to the RF signal classification task while keeping the number of trainable parameters below 3% of the total model size. The ratio $\alpha/r = 2$ is a standard initialization that has been empirically shown to provide stable training dynamics across a wide range of fine-tuning tasks. LoRA was applied to all linear layers of the model (`target_modules = “all-linear”`), yielding approximately 7.7M trainable parameters out of 256M total, corresponding to 2.93% of the full model.

2.6.5 Label Masking

A critical implementation detail in autoregressive fine-tuning is the correct masking of the loss computation. The cross-entropy loss must be computed exclusively over the tokens corresponding to the model’s answer, and not over the system prompt, the user query, or the image patch tokens. Failure to apply this masking correctly results in the model being trained to predict its own input, which leads to artificially low training loss without genuine task learning.

In SmolVLM-256M, the full tokenized input sequence has the structure:

$$\underbrace{[\text{system tokens}]}_{\text{masked}} \quad \underbrace{[\text{image patch tokens}]}_{\text{masked}} \quad \underbrace{[\text{query tokens}]}_{\text{masked}} \quad \underbrace{[\text{answer tokens}]}_{\text{unmasked}} \quad (31)$$

The image patch tokens occupy a substantial portion of the sequence — approximately 1,100 tokens for a 256×256 input — and must be explicitly masked. The **Assistant:** turn boundary was identified by locating the token pattern corresponding to the string "**Assistant:**" (token IDs [9519, 9531, 42]) within the full tokenized sequence, and all tokens preceding this boundary were assigned a label of -100 , which is the standard PyTorch convention for ignoring a position in the cross-entropy loss. Only the answer tokens and the end-of-utterance token were left unmasked.

2.6.6 Training Configuration

The model was fine-tuned using the Hugging Face **Trainer** API [12], with the **transformers**, **datasets**, **peft**, and **torch** libraries. Training was conducted on a single NVIDIA A100 GPU provided by Google Colab. Due to computational constraints, a subset of the full dataset was used: 48,000 training samples (corresponding to 9600 unique constellation images $\times$ 5 questions), 5,000 validation samples (1,000 images), and 10,000 test samples (2,000 images). The training hyperparameters are summarized in Table 7.

Table 7: SmolVLM-256M fine-tuning configuration

Hyperparameter	Value
Base model	SmolVLM-256M-Instruct
LoRA rank r	16
LoRA scaling α	32
LoRA target modules	all-linear
Trainable parameters	$\sim 7.7\text{M}$
Optimizer	AdamW
Learning rate	1×10^{-4}
Warmup ratio	0.05
Batch size (per device)	16
Gradient accumulation	2 (effective batch: 32)
Precision	bfloat16
Epochs	2
Training samples	48,000
Validation samples	5,000
Hardware	NVIDIA A100 (Google Colab)

2.6.7 Training Results

The model was trained for two epochs. The training and validation losses for each epoch are reported in Table 8. The consistent decrease in both training and validation loss across epochs, combined with the small gap between the two, confirms that the model learned generalizable features without overfitting to the training set.

Table 8: SmolVLM-256M training and validation loss per epoch

Epoch	Training Loss	Validation Loss
1	0.206940	0.224661
2	0.169408	0.181245

The relatively rapid convergence — with training loss dropping from approximately 9.45 at step 10 (before the warmup phase completed) to 0.20694 by the end of epoch 1 — is consistent with the established behavior of LoRA fine-tuning, where the pretrained model’s representations require only modest adaptation to the new domain. The epoch-over-epoch improvement in training loss and validation loss suggests that further training epochs would probably yield favourable returns, but two epochs are a computationally justified stopping point for the available hardware budget.

2.7 Detailed Performance Analysis of the Fine-Tuned VLM

The evaluation of the fine-tuned SmolVLM-256M on the test set of 2,000 images yields an overall multi-task accuracy of **67.36%**. This score reflects a highly non-uniform performance across the five classification objectives. The breakdown reveals that the Vision Transformer (ViT) backbone excels at aggregating global, macroscopic image context but struggles with micro-spatial geometric precision.

2.7.1 External Interference: Jamming Severity

The detection and classification of jamming interference is the model’s strongest capability, reaching an exceptional accuracy of **92.65%**.

- **Detection Precision:** The model achieves perfect or near-perfect recall for the Severe (1.00) and Medium (0.98) jamming levels. Jamming typically manifests as widespread circular halos or broad textural changes across the polar plot. The ViT’s patch-based self-attention mechanism is highly proficient at capturing these large-scale spatial disruptions.
- **Sensitivity Thresholds:** Minor classification leakage occurs primarily between the Mild (recall: 0.84) and None (recall: 0.90) categories.

2.7.2 Signal-to-Noise Ratio (SNR) Range Estimation

The SNR range estimation task achieved a highly robust accuracy of **89.20%**.

- **Entropy Recognition:** The model shows outstanding performance at the lower extremes, successfully identifying Very Low SNR with a recall of 0.98 and precision of 0.96. The ViT effectively evaluates the overall spatial dispersion and pixel density of the noise clouds, interpreting the high-entropy scatter as severe signal degradation.
- **Clarity Saturation:** A noticeable performance dip occurs in the Very High category (recall: 0.77). For exceptionally clean signals, the differences in spatial variance become remarkably subtle. The 32×32 pixel patch tokens of the vision encoder lack the granular resolution required to distinguish these minute variance shifts, occasionally leading the model to conservatively predict the SNR as High.

2.7.3 Modulation Type Identification

The model achieves a moderate accuracy of **63.05%** in identifying the modulation scheme, illustrating the complexities of purely spatial demodulation.

- **Low-Order Isolation:** The model successfully identifies simpler alphabets, achieving high recall for BPSK (0.93), QPSK (0.82), and 4-HQAM (0.93). The distinct macroscopic shapes (e.g., singular lines vs. four distinct quadrants) are easily mapped by the self-attention layers.

- **High-Order Overlap:** Performance drops significantly for dense schemes like 64-QAM (recall: 0.29), 256-QAM (recall: 0.45), and 64-APSK (recall: 0.27). In a blind scenario, the overlapping Gaussian distributions of high-order constellations form ambiguous, visually identical noise blobs. Without explicit reference coordinates to anchor the cluster centers, the visual tokens cannot resolve the exact grid density, leading to inter-class confusion.

2.7.4 Transceiver Impairments: I/Q Imbalance Severity

The estimation of **I/Q Imbalance** proves to be a significant challenge, yielding an accuracy of only **48.10%**.

- **Severe Detection:** The model demonstrates meaningful discriminatory power primarily for Severe distortions (recall: 0.72). At this level, the non-uniform elliptical compression is extreme enough to alter the global topological footprint of the signal clouds.
- **Axis-Dependency Failure:** The network catastrophically fails to identify Mild (recall: 0.26) and Medium (recall: 0.37) imbalances. Detecting slight asymmetrical compression requires precise axis-dependent measurements. The patch embeddings of the ViT prioritize summarized global context over exact, sub-pixel coordinate mapping, effectively ignoring these subtle geometric stretches.

2.7.5 Transceiver Impairments: Phase Noise Severity

The classification of **Phase Noise** represents the most severe failure mode for the architecture, with an accuracy of **43.80%**.

- **Resolution Limitations:** The model struggles heavily with Mild (recall: 0.28) and None (recall: 0.34) categories. Phase noise introduces a continuous rotational smearing. Because the visual encoder slices the image into fixed patches, micro-rotational jitters are smoothed out and absorbed into a single token’s embedding during the linear projection phase.
- **Macroscopic Tracking:** The recall recovers slightly for Severe phase noise (0.72), where the rotational variance is large enough to create massive, unmistakable arcs across multiple image patches. Ultimately, this confirms that vision-language models lack the native geometric strictness required for fine-grained phase tracking.

2.7.6 Visual Forensics: Confusion Matrix Interpretation

While the overall accuracy metrics provide a numerical summary, the confusion matrices reveal the internal logic—and the specific failures—of the Vision-Language Model’s interpretive process in a strictly blind scenario. The following paragraphs analyze how the patch-based self-attention mechanism processes geometric and statistical RF features without the aid of ideal reference anchors.

Modulation Identification: The Limits of Geometric Resolution The confusion matrix for modulation (Figure 20) illustrates a clear divide between the model’s ability to recognize fundamental shapes and its struggle with high-density overlaps.

- **Low-Order Isolation:** The model successfully identifies simpler alphabets with distinct macroscopic topologies. It achieves high true-positive rates for schemes like BPSK (119 correct), QPSK (109 correct), and 4-HQAM (111 correct), proving that the ViT effectively captures distinct quadrant formations and singular signal lines.

- **High-Order Fragmentation:** The diagonal breaks down significantly for dense, complex constellations. Without spatial anchors to define cluster centers, the model struggles to differentiate between overlapping Gaussian distributions. For example, 64-QAM is frequently confused with 256-QAM, and 128-APSK exhibits significant scatter across neighboring APSK and QAM families, confirming that the model’s visual tokens cannot resolve precise grid densities in a blind context.

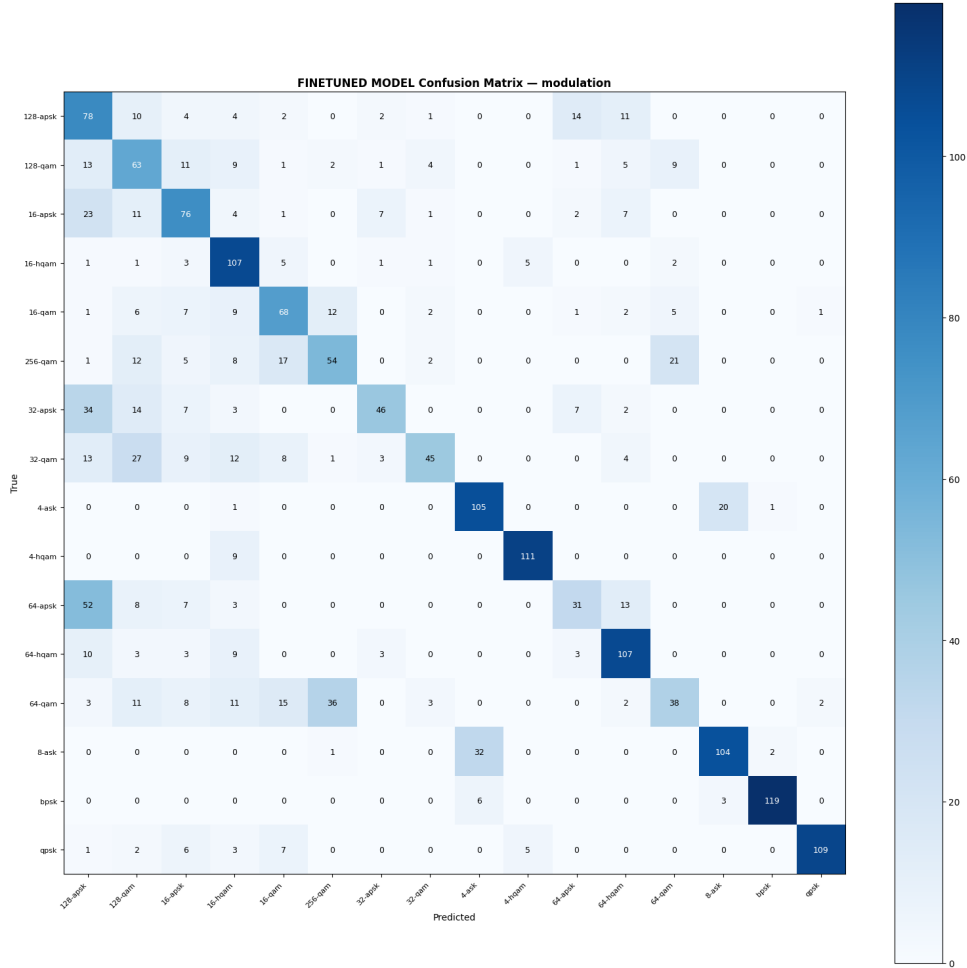


Figure 20: Confusion Matrix for Modulation. Strong predictive accuracy is isolated to low-order schemes, while high-order alphabets suffer from severe inter-class leakage due to overlapping spatial distributions.

Jamming Detection: Macroscopic Texture Recognition The jamming matrix (Figure 21) highlights the model’s exceptional capability in detecting widespread structural interference.

- **Interference Certainty:** The model demonstrates outstanding precision for high-power interference, correctly identifying 493 cases of **severe** jamming and 483 cases of **medium** jamming with almost zero leakage into non-adjacent classes. The model easily perceives the large-scale "halo" effect caused by the jammer.
- **The Thermal Noise Boundary:** The only notable confusion occurs at the threshold of detection. The matrix reveals that 69 instances of **mild** jamming were predicted as **none**, and 42 instances of **none** were predicted as **mild**. This indicates that when the Jammer-to-Noise Ratio (JNR) is extremely low, the visual signature merges with the standard thermal noise floor, causing slight ambiguity.

FINETUNED MODEL Confusion Matrix — jamming

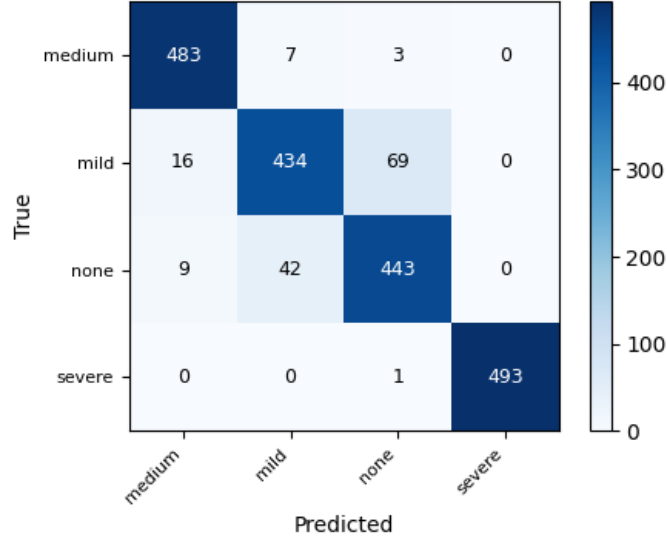


Figure 21: Confusion Matrix for Jamming Severity. The strong diagonal confirms the VLM’s proficiency in recognizing macroscopic interference halos.

SNR Range: The Entropy Gradient The SNR matrix (Figure 22) visualizes how the model evaluates overall spatial dispersion and pixel density.

- **High-Entropy Success:** The network effectively isolates states of extreme signal degradation, successfully classifying 474 cases of **very_low** SNR with negligible leakage. It recognizes the complete collapse of the constellation into a uniform Gaussian cloud.
- **Adjacent Class Leakage:** Errors are strictly confined to neighboring entropy tiers, which is typical for continuous variables. For instance, 50 **very_high** cases were labeled as **high**. This reveals that for exceptionally clean signals, the minute differences in point-spread variance fall below the resolution limits of the model’s 32×32 pixel patches.

Transceiver Impairments: Micro-Spatial Resolution Failure The matrices for I/Q Imbalance (Figure 23) and Phase Noise (Figure 24) depict a severe breakdown in discriminatory power, exposing the limitations of autoregressive vision models in tasks requiring sub-pixel accuracy.

- **Asymmetrical Blindness (I/Q Imbalance):** The matrix for I/Q Imbalance exhibits heavy dispersion. Notably, 150 cases of **mild** distortion were incorrectly predicted as **none**. Because I/Q imbalance introduces an axis-dependent elliptical compression, the lack of a central geometric anchor prevents the model from measuring this subtle non-uniform stretching, leading to high false-negative rates.
- **Rotational Smoothing (Phase Noise):** The Phase Noise matrix shows a near-catastrophic spread across all categories. The **mild** class is essentially guessed at random (with 133 predicted as **none** and 133 correctly predicted). This confirms that the patch-based embedding process of the Vision Transformer acts as a spatial low-pass filter; the subtle, continuous rotational jitter is "smoothed out" and lost within the token representations, making the network virtually blind to micro-rotational phase variance.

FINETUNED MODEL Confusion Matrix — snr_range

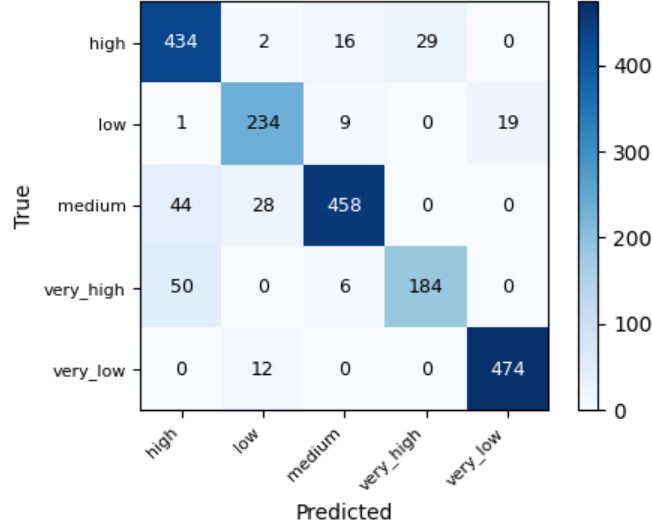


Figure 22: Confusion Matrix for SNR Range. Misclassifications are strictly limited to adjacent tiers, demonstrating a solid understanding of generalized spatial variance.

FINETUNED MODEL Confusion Matrix — iq_imbalance

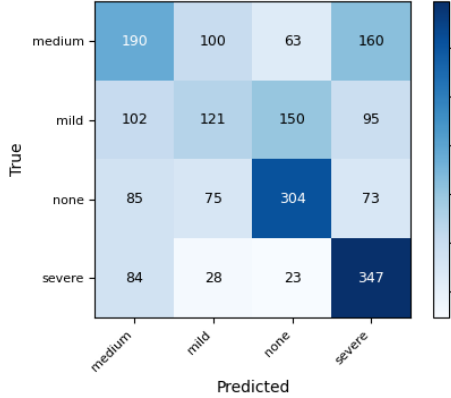


Figure 23: Confusion Matrix: I/Q Imbalance. High dispersion indicates an inability to track asymmetrical compression without geometric anchors.

FINETUNED MODEL Confusion Matrix — phase_noise

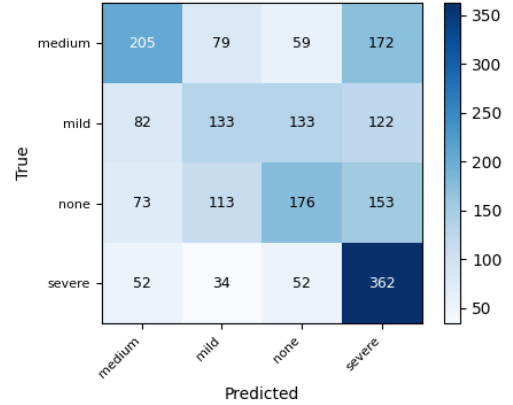


Figure 24: Confusion Matrix: Phase Noise. The near-random distribution confirms that patch-based VLMs fail to capture sub-pixel rotational smearing.

2.7.7 Robustness and Generalization Testing: Accuracy vs. SNR Analysis

To rigorously evaluate the reliability of the fine-tuned SmolVLM under conditions of dataset shift, an Out-Of-Distribution (OOD) robustness test was conducted. A dedicated set of 400 novel images was generated using intermediate, continuous parameter values that the network did not encounter during training. This included unseen SNR tiers (e.g., 3, 7, 12, 18 dB) and recalibrated continuous impairment ratios. By directly comparing the in-distribution test results against this OOD set, we can determine whether the Vision-Language Model learned generalized spatial topologies or merely overfit to discrete visual templates.

The comparative analysis reveals an impressive structural stability across the majority of the classification tasks:

- **Modulation Classification (63.05% to 63.25%):** The performance remains remarkably stable. This statistical equivalence indicates that the Vision Transformer’s self-attention mechanism successfully generalized the fundamental topological signatures of the constellations. It proves the model is immune to the intermediate noise boundaries introduced in the OOD set, recognizing the core shapes rather than memorizing specific noise footprints.
- **Jamming Detection (92.65% to 92.25%):** The model exhibits near-perfect resilience. Because jamming introduces macroscopic, global interference halos, shifting the exact jamming power to continuous unseen levels does not alter the overarching visual texture recognized by the patch embeddings.
- **I/Q Imbalance Severity (48.10% to 48.25%):** The accuracy shows no degradation. While the overall score reflects the inherent difficulty of measuring asymmetrical compression without spatial anchors, the stability proves that the model’s partial understanding of this distortion is generalized and not merely overfit to the training set’s specific elliptical ratios.
- **Phase Noise Severity (43.80% to 44.50%):** Similarly, phase noise performance remains virtually identical. The model maintains its baseline ability to detect macroscopic rotational smearing without suffering a catastrophic collapse when exposed to novel, continuous phase variances, although the performance is still very low.
- **SNR Range Estimation (89.20% to 76.00%):** This is the only task that experiences a noticeable performance drop. This degradation is mathematically expected when dealing with continuous variables: introducing intermediate SNR values (e.g., 7 dB or 12 dB) forces the network to evaluate spatial dispersions that lie exactly on the ambiguous decision boundaries of the predefined severity categories, naturally increasing thresholding confusion.

This behavioral stability is further illuminated by visualizing the model’s performance across the continuous SNR spectrum.

As observed in Figure 25, the trajectories for tasks relying on global geometric structure (such as Modulation and Jamming) maintain consistent, predictable curves despite the unseen SNR injections. It is important to note that in the case of the Fine-Tuned SmolVLM, in contrast to traditional CNN architectures, shows a much more expected dependency of high SNR values to correctly predict most of the given tasks.

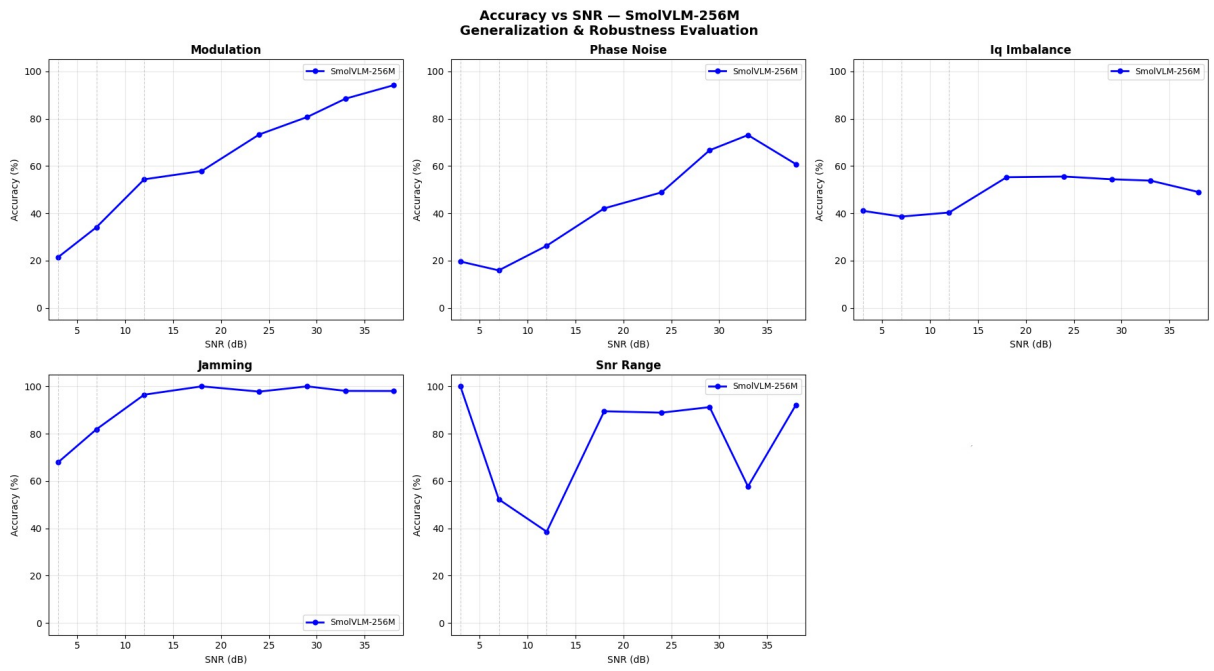


Figure 25: SmolVLM Accuracy vs. SNR across the five classification tasks. The dashed vertical lines indicate the unseen OOD SNR values (3, 7, 12 dB, etc.).

3 Comparative Analysis

This research has explored a diverse spectrum of Artificial Intelligence architectures—ranging from lightweight convolutional neural networks to massive vision-language foundation models—applied to the domain of RF signal classification. This final section synthesizes the experimental findings, analyzing each model’s strengths and weaknesses, outlining the strategic trade-offs between computational efficiency and classification robustness.

3.1 Comparative Performance Analysis Across Architectures

To determine the most effective paradigm for strictly blind RF constellation analysis, the performance of the three distinct architectures—LeNet, ResNet-18, and the fine-tuned SmolVLM—was directly compared. The results, summarized in Table 9, reveal distinct architectural strengths and vulnerabilities when processing spatial topologies without the aid of ideal reference anchors.

Table 9: Comparative Accuracy Across All Evaluated Architectures (Test Set)

Classification Task	LeNet	ResNet-18	SmolVLM
Modulation Classification	64.12%	71.11%	63.05%
Jamming Severity	89.16%	92.33%	92.65%
SNR Range Estimation	80.04%	88.22%	89.20%
I/Q Imbalance Severity	55.66%	73.68%	48.10%
Phase Noise Severity	33.93%	45.52%	43.80%

Macroscopic Feature Extraction: Jamming and SNR For tasks involving global structural disruptions and spatial variance, the deeper models demonstrate near-parity and exceptional performance. SmolVLM achieves the highest accuracy in Jamming Detection (92.65%) and SNR Range Estimation (89.20%), narrowly outperforming ResNet-18 (92.33% and 88.22%, respectively). This indicates that the Vision Transformer’s self-attention mechanism is highly adept at capturing widespread interference halos and calculating generalized image entropy. Conversely, the shallow architecture of LeNet (80.04% for SNR) struggles slightly more with continuous variance estimation, though it remains robust in detecting macroscopic jamming (89.16%).

Geometric Resolution: Modulation Classification When identifying precise structural topologies (Modulation), ResNet-18 establishes a clear superiority, achieving 71.11% accuracy compared to LeNet’s 64.12% and SmolVLM’s 63.05%. While the VLM and LeNet can successfully isolate simpler, low-order alphabets, they falter when distinguishing dense, overlapping probability distributions (e.g., high-order QAMs). The deep, hierarchical spatial filters and residual connections of ResNet-18 allow it to extract finer cluster boundaries from the noise clouds, giving it a definitive edge in pure spatial demodulation.

The Micro-Spatial Divide: I/Q Imbalance The most dramatic performance gap between the architectures emerges in the characterization of I/Q Imbalance. ResNet-18 dominates this task with an accuracy of 73.68%, whereas LeNet achieves 55.66% and SmolVLM falls to 48.10%. I/Q Imbalance requires tracking asymmetrical, axis-dependent compression. ResNet-18 successfully preserves these subtle elliptical gradients through its deep identity mappings. In contrast, LeNet destroys this geometric precision via aggressive max-pooling, and the VLM’s patch-based embeddings act as a spatial low-pass filter, effectively blinding the self-attention mechanism to sub-pixel asymmetrical stretches.

The Universal Bottleneck: Phase Noise Despite the differing computational philosophies, Phase Noise remains the most formidable challenge across all three architectures. ResNet-18 (45.52%) and SmolVLM (43.80%) perform similarly, offering a modest improvement over the shallow LeNet (33.93%). This collective struggle confirms that continuous rotational smearing is inherently difficult to quantify without an exact spatial anchor. Whether an architecture relies on deep convolutional kernels or global self-attention patches, determining micro-rotational offsets in purely visual representations of RF signals presents a universal bottleneck in strictly blind scenarios.

3.2 Computational Cost and Trade-offs

The selection of an architecture for a communication system is dictated by the balance between accuracy and resource availability. Table 10 summarizes the key metrics derived from our study.

Table 10: Comprehensive Comparison of Evaluated Architectures			
Metric	LeNet	ResNet-18	SmolVLM
Parameters	~60,000	~11.7 Million	~256 Million
Inference Latency	Ultra-Low	Medium	Very High
Robustness	High	High	High

Based on the performance profiles, we identify three primary use cases for these technologies:

1. **Edge-Level Signal Identification:** For real-time deployment like Software Defined Radios (SDRs), **LeNet** is the only viable candidate. Its low parameter count allows for high-throughput processing, enabling the receiver to adapt its demodulation parameters in milliseconds. If the task requires the absolute lowest inference latency, the LeNet architecture reigns supreme whilst also showcasing an impressive amount of accuracy considering how lightweight it is.
2. **High SNR environments:** In 2.7.7, while analyzing the Accuracy-SNR graphs an interesting result was observed. The fine-tuned SmolVLM shows significant increase of performance proportionally to the increase of SNR values, almost linear in some cases. Specifically we can deduct from the graphs that the model approaches near perfect accuracy in high SNR environments when it comes to Modulation classification and Jamming severity classification, whilst also showcasing massive improvements in Phase Noise severity classification ($\sim 70\%$ in high SNR environments). Thus it is reasonable to assume that signal analysis can heavily benefit from VLM implementations in Low-Noise scenarios.
3. **Multi-Impairment Forensic Analysis:** The overall performance winner is the Resnet-18 architecture. It showcases exceptional performance whilst also having lower Computational Cost to VLM architectures. When the objective is to disentangle overlapping impairments (e.g., separating Phase Noise from I/Q Imbalance), deep residual connections are mandatory. **ResNet-18** represents the "sweet spot" for high-fidelity impairment estimation without the extreme overhead of a Transformer.

4 Concluding Remarks and Future Work

This work confirms that while macroscopic features like SNR severity and jamming can be classified by nearly any modern AI architecture, the estimation of fine-grained transceiver impairments requires a sophisticated balance of depth and spatial resolution.

The transition from LeNet to ResNet demonstrated the value of Residual Learning in preserving signal integrity, while the introduction of the SmolVLM highlighted the value of Visual Language Models in generalizing to unseen RF High-SNR environments. Future research should focus on "Lightweight Transformers" or hybrid CNN-Attention architectures, aiming to combine VLM's accuracy in Low-Noise circumstances at CNN-level inference speeds, ultimately paving the way for truly intelligent and adaptive physical layers in next-generation (6G) communication systems.

References

- [1] ETSI, “EN 302 307: Digital Video Broadcasting (DVB); Second Generation Framing Structure for Broadband Satellite Applications,” ETSI, Tech. Rep., 2009.
- [2] T. K. Oikonomou, N. G. Evgenidis, D. G. Nixarlidis, D. Tyrovolas, S. A. Tegos, P. D. Diamantoulakis, P. G. Sarigiannidis, and G. K. Karagiannidis, “Cnn-based automatic modulation classification under phase imperfections,” *IEEE Wireless Communications Letters*, vol. 13, no. 5, pp. 1508–1512, 2024.
- [3] G. K. Karagainnidis and K. N. Pappi, *Telecommunications Systems*, 4th ed. Tziola, 2011.
- [4] Y. LeCun *et al.*, “Gradient-Based Learning Applied to Document Recognition,” *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
- [5] A. Paszke *et al.*, “PyTorch: An Imperative Style, High-Performance Deep Learning Library,” *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.
- [6] K. He *et al.*, “Deep Residual Learning for Image Recognition,” *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.
- [7] H. Zou, Y. Tian, B. Wang, L. Bariah, S. Lasaulce, C. Huang, and M. Debbah, “Rf-gpt: Teaching ai to see the wireless world,” 2026. [Online]. Available: <https://arxiv.org/abs/2602.14833>
- [8] A. Dosovitskiy *et al.*, “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale,” *International Conference on Learning Representations (ICLR)*, 2021.
- [9] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [10] Hugging Face, “SmolVLM-256M-Instruct,” <https://huggingface.co/HuggingFaceTB/SmolVLM-256M-Instruct>, 2024.
- [11] E. J. Hu *et al.*, “LoRA: Low-Rank Adaptation of Large Language Models,” *International Conference on Learning Representations (ICLR)*, 2022.
- [12] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. Le Scao, S. Gugger, M. Drame, Q. Lhoest, and A. Rush, “Transformers: State-of-the-art natural language processing,” in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, Q. Liu and D. Schlangen, Eds. Online: Association for Computational Linguistics, Oct. 2020, pp. 38–45. [Online]. Available: <https://aclanthology.org/2020.emnlp-demos.6/>